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# UNION PUBLIC SERVICE COMMISSION

## COMBINED DEFENCE SERVICES (CDS) EXAMINATION

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## SECTION : NUMBER SYSTEM & DIVISIBILITY (109 QUESTIONS)

**1.** A number when divided by 7 leaves a remainder 3, and the resulting quotient when divided by 11 leaves a remainder 6. What is the remainder when the same number is divided by 77? [CDS (I) 2009 Q16]

(a) 35 (b) 40 (c) 45 (d) 50

**2.** If the same number (which leaves remainder 3 when divided by 7 and quotient leaves remainder 6 when divided by 11) is divided by 11, it leaves a remainder m, and the resulting quotient when divided by 7 leaves a remainder n. What are the values of m and n respectively? [CDS (I) 2009 Q17]

(a) 1 and 4 (b) 4 and 1 (c) 3 and 6 (d) 6 and 3

**3.** The number of pairs (x, y) where x, y are integers satisfying the equation 21x+48y=5is [CDS (I) 2009 Q18]

(a) Zero (b) One (c) Two (d) Infinity

**4.** A is a set of positive integers such that when divided by 2, 3, 4, 5 and the remainder l, 2, and 5 respec- tively. How many integers between 0 and 100 belong to the set A ? [CDS (I) 2009 Q22]

(a) No integer (b) One (c) Two (d) Three

**5.** What is the least perfect square which is divisible by 3, 4, 5, 6 and 7? [CDS (I) 2009 Q26]

(a) 1764 (b) 17640 (c) 44100 (d) 176400

**6.** Let N be the greatest number that will divide 600, 631 and 724, leaving the same remainder. What is the value of N? [CDS (I) 2009 Q41]

(a) 41 (b) 37 (c) 33 (d) 31

**7.** Let N be a 5-digit number. When N is divided by 6, 12, 15, 24 it leaves respectively 2, 8, 11, 20 as remainders. What is the greatest value of N ? [CDS (I) 2009 Q98]

(a) 99960 (b) 99956 (c) 99950 (d) 99946

**8.** A sum of money compounded annually doubles itself in 5 years. In how many years will it become four times of itself? [CDS (II) 2009 Q12]

(a) 10 years (b) 12 years (c) 15 years (d) 20 years 11

**9.** Three planets revolve round the sun once in 200, 250, days respectively in their own orbits. When do they all come relatively to the same position as at a certain point of time in their orbits ? [CDS (II) 2009 Q15]

(a) After 3000 days (b) After 2000 days (c) After 1500 days (d) After 1200 days

**10.** All odd prime numbers upto 110 are multiplied together. What is the unit digit in this product ? [CDS (I) 2010 Q3]

(a) 0 (b) 3 (c) 5 (d) None of the above

**11.** What are distinct prime mctors of the number 26381 ? [CDS (II) 2010 Q1]

(a) 29, 17, (b) 31, 17, (c) 19, 37, (d) 23, 31,

**12.** What is the sum of all the divisors of 256 ? [CDS (I) 2011 Q5]

(a) 511 (b) 510 (c) 256 (d) 255 A —

**13.** What would be the maximum value of Q in the equation 5P9+ 3R7 + 2Q8= 1114? [CDS (II) 2012 Q6]

(a) x = (b) x = (c) 330 (d) 340

**14.** In an examination, a student was asked to divide a certain number by [CDS (II) 2012 Q7]

(a) x = (b) x = (c) 330 (d) 340

**15.** By mistake he multiplied it by 8 and got the answer 2016 more than the correct answer. What was, the number ? [CDS (II) 2012 Q8]

(a) 252 (b) 256 (c) 258 (d) 260

**16.** A man undertakes to do a certain work in 150 days. He employs 200 men. He finds that only a quarter of the work is done in 50 days. How many additional men should he employ so that the whole work is finished in time ? [CDS (II) 2012 Q11]

(a) 75 (b) 85 (c) 100 (d) 120

**17.** The annual income of a person decreases by T 64 if the annual rate of interest decreases from 4% to 3-75%. What is his original annual income ? [CDS (II) 2012 Q15]

(a) ^ 24,000 (b) r 25,000 (c) ^ 25,600 (d) ^ 24,600

**18.** The average of the temperatures recorded at noontime from Monday to Sunday is 31 oc. Ifthe lowest tempera- ture recorded is 300C, then what is the maximum of temperature that is possible to record at noontime on any one of the days [CDS (I) 2013 Q4]

(a) 340C (b) ,350C (c) 360C (d) 370C

**19.** Which one of the following has least number of divisors ? [CDS (II) 2013 Q4]

(a) 88 (b) 91 (c) 96 (d) 99

**20.** What are the factors of [CDS (I) 2014 Q3]

(a) (x —2y—4) and (x —2Y+2) (b) (x —y and (x —4 y + 4) (c) (x —y and (x —4 y —4) (d) (x (x +2Y+2) 26k /

**21.** If X takes 45 minutes more than Y to reach the destination, then what is the time taken by Y to reach the destination ? [CDS (I) 2014 Q9]

(a) 135 minutes (b) 150 minutes (c) 180 minutes (d) 225 minutes

**22.** The price of an article X increases by 20% every year and price of article Y increases by 10% every year. In the year 2010, the price of article X was C 5000 and price of article Y was 0000. In which year the difference in their prices exceeded 0000 for the first time ? [CDS (I) 2015 Q2]

(a) 2012 (b) 2013 (c) 2014 (d) 2015

**23.** What is the average marks of all the candidates? [CDS (I) 2015 Q6]

(a) 758 (b) 75-8 (c) 760. (d) 768'.

**24.** What is the sum of their reciprocals ? [CDS (I) 2015 Q9]

(a) 2 (b) 2 (c) 2 (d) 2-562 + b2

*— SPACE FOR ROUGH WORK —*

**25.** How many zeros are there in the product 150 x 249 x 348* x 501 ? [CDS (I) 2016 Q7]
(a) 262 (b) 261 (c) 246 (d) 235

**26.** The number of three digit numbers (all digits are different) which are divisible by 7 and also divisible by 7 on reversing the order of the digits, is [CDS (II) 2016 Q45]
(a) Six (b) Five (c) Four (d) Three

**27.** What is the sum of the largest and the smallest 4-digit numbers made by using single digit prime numbers (without repetition) ? [CDS (II) 2016 Q58]
(a) 7887 (b) 7997 (c) 8998 (d) 9889

**28.** If a Star figure is formed by elongating the sides of a regular pentagon, then the measure of each angle at the angular points of the star figure is [CDS (I) 2017 Q12]
(a) 360 (b) 350 (c) 320 (d) 300

**29.** When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third mysition is [CDS (I) 2018 Q4]
(a) 1000m (b) 800m (c) 750m (d) 600 m

**30.** If the average of 9 consecutive positive integers is 55, then what is the largest integer ? [CDS (I) 2019 Q100]
(a) 57 (b) 58 (c) 59 (d) 60

**31.** If the average marks of the complete class is 72, then what is the average marks of the girls ? [CDS (I) 2020 Q11]
(a) 64 (b) 65 (c) 68 (d) 74

**32.** A cricketer has a certain average of 10 innings. In the eleventh inning he scored 108 runs, thereby increasing his average by 6 runs. What is his new average ? [CDS (II) 2021 Q4]
(a) 42 (b) 47 (c) 48 (d) 60

**33.** A wheel makes 360 revolutions in one minute. What is the number of radians it tums in one second ? [CDS (II) 2021 Q9]
(a) 10 (b) 16i (c) 36 (d) 27

**34.** Two circles touch internally. The sum of their areas is 1367t cm2 and distance between their centres is 4 What are the [CDS (I) 2022 Q4]
(a) +1 (b) + (c) l (d) 266+1

**35.** The area of a rectangle whose length is 5 units more than twice its width is 75 square units. What is its width ? [CDS (I) 2022 Q9]
(a) 3 units (b) 5 units (c) 7 units (d) 10 units

**36.** Which product is produced least over the years ? [CDS (II) 2023 Q10]
(a) Type (b) Type (c) Type (d) Type V ,'SOA

**37.** In Which one of the following pairs of years, the difference in total number of tablets produced between them is m inimum? [CDS (II) 2023 Q11]
(a) (2003, 2005) (b) (2001, 2005) (c) g003, 2004) (d) (2000, 2002)

**38.** Ifthe length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time ? [CDS (II) 2025 Q10]
(a) 150 (b) 300 (c) 450 (d) 600

**39.** If the same number when divided by 11 leaves a remainder m and the resulting quotient when divided by 7 leaves a remainder n. What are the values Of m and n respectively ? [CDS (I) 2009 Q17]
(a) I and 4 (b) 4 and 1 (c) 3 and 6 (d) 6 and 3 4

**40.** The sum of the numbers is not divisible by(X+Y + Z). [CDS (I) 2009 Q23]
(a) ab 1 (b) abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to
(c) abc (d) xyz

**41.** The sum of the numbers is divisible by [CDS (I) 2009 Q24]
(a) ab 1 (b) abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to
(c) abc (d) xyz

**42.** What is the remainder when 111222 + 222333 + 333444 is divided by 5 ? -3 S. What are the last digits in the multiplication of 4321012345 x 98766789 ? 6. p varies directly as (x-2 + Y2 + zz). When 3, thenp—70. What is the value of p when x —1, y = 1, [CDS (I) 2009 Q29]
(a) 100 (b) 125 (c) 135 (d) 140

**43.** Let p and q be two natural numbers such that (p + is divisible by [CDS (I) 2009 Q38]
(a) 12 21. What is c2 + bc—a2 —ab equal to? (b) -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 —4x— IO equal to? If equal to ?
(c) 2p (d) then what is then what is x 22. If ab ba , then what is equal to?

**44.** The remainder when 3x^ + kx^ + 5x - 6 is divided by (x + 1) is - 7. What is the value of k ? [CDS (II) 2009 Q100]
(a) - 14 (b) 14 (c) - 7 (d) 7

**45.** o 50 mt•, 503?R 4 4 3 3 8 27 4 IO. t, 3x+l+ 3r+2 (b) 13 (c) 39 (d) 117 PIXI-T-MTK - A 11. Q x:i7ÄT, [CDS (I) 2010 Q7]
(a) 81 (b) 13
(c) 39 (d) 117 PIXI-T-MTK - A 11. Q x:i7ÄT,

**46.** Unit in is 7. of the square of two is always divisi- by [CDS (I) 2011 Q10]
(a) be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ? (b) 80
(c) 60 36 • (d) 24 14. WEch Of

*— SPACE FOR ROUGH WORK —*

**47.** 20% t 30% t, 9 32 43 24. 19 Q 182 I 175 I [CDS (II) 2011 Q7]

(a) 2975 (b) 3094

(c) 3101 (d) 3269 25. 48 x:nfq 160 qqfq 2 3 4 27. t I PQ-3cm, QR=9an W-10-5cm t,

**48.** A merchant commences with a certain capital and gains annually at the rate of 25%. At the end of 3 years he has 10,000. What is the original amount that the merchant invested ? [CDS (II) 2012 Q14]

(a) ?= 5,120 (b) ^ 5,210 (c) ^ 5,350 (d) ^ 5,500

**49.** Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? [CDS (I) 2013 Q8]

(a) Its lia ixuide O (b) Its lies on the triülgle

(c) It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? (d) AB<RX

**50.** If = l, then What is value of + 3iniOx + + ? •13 [CDS (I) 2013 Q9]

(a) Its lia ixuide O (b) Its lies on the triülgle

(c) It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? (d) AB<RX

**51.** How pairs Of ze such that åe of thü Nuar— 35 ? [CDS (I) 2013 Q14]

(a) -1.2 (b) -1,-2 4 90. How pairs Of 15 ze such that åe of thü Nuar— 35 ? 91. If

(c) 21 (d) 19 value of x2+5x+ O 94. mtio of

**52.** If are the roots Of this equation? 2' 2b [ P.T.O. [CDS (I) 2014 Q13]

(a) 12 21. What is c2 + bc—a2 —ab equal to? (b) -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 —4x— IO equal to? If equal to ?

(c) 2p (d) then what is then what is x 22. If ab ba , then what is equal to?

**53.** Le g nd b be real + = 32 50 + b) 7 (d) be 2 one Of the 61%wing ? [CDS (I) 2015 Q1]

(a) be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ? (b) 80

(c) 60 36 • (d) 24 14. WEch Of

**54.** Number of times a }xrson correctly hits a target in ten attempts Which of the following in respect of the above is correct ? [CDS (I) 2015 Q12]

(a) I is a continuous variable, 2 is not a variable and 3 is a discrete variable (b) 1 is a continuous variable, 2 and 3 are discrete variables

(c) I and 3 are discrete variables and 2 is not a variable (d) •l, 2 and 3 all are discrete variables Consider the followiög •statements :

**55.** Which of the above statements is/are correct? [CDS (I) 2017 Q5]

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither nor 2 Which one is one of the factors of (a) (c) / 48A

**56.** If O is an acute angle and sinecos0= 2cos30 — 1-5cosO, then what is sine equal to ? 6-1 4 1-6 (b) 4 4 4 [CDS (I) 2017 Q14]

(a) tan29 (b) sin2e COS 9 38. is equal to I — sine tang— I tang +1 1 + sine

(c) coso tano + I tang — I (d) sin 6 39. If tan

**57.** If 4, Where O then what is cosece equal to? (b) 3/2 5/3 [CDS (I) 2019 Q5]

(a) 3/2 5/3 33. If 2 tane=sec29 —2, where R 12, then what is cote equal to? (b) v'ä-l

(c) 0+2 I — sin 34. What is (d) mane

**58.** What is the value of sin2e cos29 (s«2e + cosæ2e) equal to ? sinle [CDS (II) 2020 Q4]

(a) 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e (b) 2h

(c) Htana•tanß (d) h cota•cOtß

**59.** If 64 64 - 16 where O O — , 2 then what is the value of + coto ? [CDS (II) 2020 Q5]

(a) 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e (b) 2h

(c) Htana•tanß (d) h cota•cOtß

**60.** If cosece — Cote — m and sece — tane n, then what is cosece + sece qual to ? (d) [CDS (II) 2020 Q6]

(a) 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e (b) 2h

(c) Htana•tanß (d) h cota•cOtß

**61.** If 2 tane=sec29 —2, where R 12, then what is cote equal to? (c) v'ä-l (d) 0+2 [CDS (II) 2020 Q11]

(a) 3/2 5/3 33. If 2 tane=sec29 —2, where R 12, then what is cote equal to? (b) v'ä-l

(c) 0+2 I — sin 34. What is (d) mane

**62.** Consider the following statements : [CDS (I) 2025 Q15]

(a) ab 1 (b) abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to

(c) abc (d) xyz

**63.** a, — + a, —a, + as is divisible by [CDS (II) 2009 Q42]

(a) I only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2 What is x

*— SPACE FOR ROUGH WORK —*

**64.** A two-digit number is such that the product of the digits is [CDS (II) 2010 Q4]
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2 only when Ware correct ? 28. Let work done by

**65.** Consider the following : I. 10+ I [CDS (I) 2015 Q14]
(a) p (b) 3 and —4 (c) —3 and—4 (d) —3 and 4

**66.** Let x and y be positive integers such that x is prime and y is composite. Which of the following statements are correct ? 1. (y —x) can be an even integer. 2. can an even integer. 3. , can be an even integer. - A of Select the correct answer using the code given below : [CDS (II) 2016 Q38]
(a) I and 2 only (b) 2 and 3 only (c) I and 3 only (d) 1, 2 and 3

**67.** S is always divisible by 3 and divisible by 9 divisible by 37 15 15 Select the corect answer using the code given below : [CDS (II) 2016 Q46]
(a) 1 only (b) 2 only (c) 1 and 2 (d) I and 3

**68.** Consider the following statements : I. Two consecutive natural numbers are always co-prime. [CDS (II) 2016 Q61]
(a) I only (b) 11 only
(c) Both land 11 (d) Neither I nor 11 29. If and then what is log10

**69.** Educational qualification of the person Continuous variable Attribute 28 Select the correct answer using the code given below • [CDS (I) 2017 Q3]
(a) I only (b) 2 only (c) Both I and 2 (d) Neither 1 nor 2

**70.** Select the correct •answer using the code given below • [CDS (I) 2017 Q13]
(a) I only (b) 2 only (c) Both I and 2 (d) Neither I nor 2 8

**71.** If X and Y work alternately starting with Y, then the piece of work will be finished in less than 5 days. Select the answer using the code given below : [CDS (I) 2018 Q15]
(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**72.** What is tanß equal to? p.T.O. [CDS (I) 2019 Q6]
(a) 11 only (b) Both I and 11
(c) Neither I nor 11 39. Which Of the is/ are the Of (d) Both I and 11

**73.** then which Of the following is/ are correct? Select the correct answer using the code given below. [CDS (I) 2020 Q3]
(a) I only (b) 2 only
(c) Both I and 2 (d) Neither I nor 2 710 — 510 is divisible by 10 (d) 11

**74.** 13y) n. (x+19y) Select the correct answer using the code given below. [CDS (II) 2020 Q12]
(a) I only (b) 11 only (c) Both 1 and 11 (d) Neither I nor 11

**75.** What is +3 equal to? (b) 1/2 For the tvo items : Let psin2 +qcos2 a qsin2 pcos2 ß=n; p m, n and q m. n. [CDS (II) 2020 Q13]
(a) 11 only (b) Both I and 11
(c) Neither I nor 11 39. Which Of the is/ are the Of (d) Both I and 11

**76.** Area of AQAB is equal to twice that of APCQ Select the correct answer using the code given below : [CDS (I) 2021 Q7]
(a) 1 only (b) 2 only (c) Both I and 2 (d) Neither 1 nor 2

**77.** Q can be a perfect square. Q is positive for all real values of x. 9 Select the answer using the code given below : [CDS (I) 2021 Q11]
(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**78.** Both the numbers may be com- posite. Select the correct answer using the code given below : [CDS (II) 2021 Q1]
(a) 1 and 2 only (b) 2 and 3 Only (c) 1 and 3 only (d) I, 2 and 3

**79.** For what integral value of x is (b) [CDS (II) 2021 Q14]
(a) Which Of the above statements is/are correct? (b) I only
(c) 2 only (d) Both I and 2

**80.** 3 2 — + I = k 2 , then which One of the is a possible expression k P — 3X+I [CDS (II) 2023 Q6]
(a) Which Of the above statements is/are correct? (b) I only
(c) 2 only (d) Both I and 2

**81.** What is bc(a — b) (a — equal to ? [CDS (II) 2023 Q7]
(a) Which Of the above statements is/are correct? (b) I only
(c) 2 only (d) Both I and 2

**82.** Consider the following statements in respect of where n is a natural number : I. p is always (Xid 11. p is a perfect square Which of the statements given above is/ are correct ? [CDS (I) 2009 Q34]
(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11 A -

**83.** Let P = N2 where N is an odd integer. What is the remainder when P is divided by 8 ? (d) Cannot be determined due to insufficient data [CDS (I) 2009 Q42]
(a) I only (b) 11 only
(c) Both land 11 (d) Neither I nor 11 29. If and then what is log10

**84.** What is the remainder when 2n is divided by 4 ? What is the last digit in the expansion of [CDS (I) 2009 Q75]
(a) I only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2 What is x

**85.** divides 675 + 335 Which of the statements given above is/ are correct ? [CDS (II) 2009 Q3]
(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

*— SPACE FOR ROUGH WORK —*

**86.** If n is a prime number greater than 5, then n' — 1 is divisible by 2400. Every square number is of the form 5n or (5n — 1) or (5n + l) where n is a whole number. Which of the above Statements is/are correct ? [CDS Math PYQ Chapter ch_1]
(a) 1 only (b) 2 only
(c) Both land 2 (d) Neither 1 nor 2 33. kx — 6, then what is the value Of

**87.** What is the remainder when 3255 is divided by 28 ? (b) 11 (c) 24 (d) 27 [CDS Math PYQ Chapter ch_1]
(a) nM — k (b) n (c) 81 (d) 105

**88.** All odd prime numbers upto 110 are multiplied together. What is the unit digit in this product ? [CDS Math PYQ Chapter ch_1]
(a) 0 (b) 3 (c) 5 (d) None of the above

**89.** appear for the examination, — of those who 20 appeared passed. If the number of registered students is 2000, the number who passed is [CDS Math PYQ Chapter ch_1]
(a) 1920 (b) 1056 (c) 1020 (d) 864

**90.** If A : B = 1 : 2, B : C = 3 : 4, C : D = 2 : 3 and D : E = 3 : 4, then what is B : E equal to ? [CDS Math PYQ Chapter ch_1]
(a) 3 :2 (b) 1: 8 (c) 3 :8 (d) 4 :1

**91.** For 0 < m < 1, which one of the following is correct ? [CDS Math PYQ Chapter ch_1]
(a) ■ logj^Q m < m^ < m < m“ ^ (b) m < m~^ < m^ < logjQ m
(c) logio ni< m < m“ ^ < m^ (d) • logj^Q m < m“ ^ < m < m^

**92.** A man undertakes to do a certain work in 150 days. He employs 200 men. He finds that only a quarter of the work is done in 50 days. How many additional men should he employ so that the whole work is finished in time ? [CDS Math PYQ Chapter ch_1]
(a) 75 (b) 85 (c) 100 (d) 120

**93.** ^ 120 is distributed among A, B and C so that A's share is ^ 20 more than B's and W 20 less than C's. What is B's share ? [CDS Math PYQ Chapter ch_1]
(a) 10 (b) 15 (c) 20 (d) 25

**94.** Ifp+q+r=0, then what is equal to? (c) 22 (d) 23 [CDS Math PYQ Chapter ch_1]
(a) -23 (b) -17 (c) 17 (d) 23 which —5k + 4

**95.** The annual income of a person decreases by T 64 if the annual rate of interest decreases from 4% to 3-75%. What is his original annual income ? [CDS Math PYQ Chapter ch_1]
(a) ^ 24,000 (b) r 25,000 (c) ^ 25,600 (d) ^ 24,600

**96.** What is 25 16 16 19 4 4 [CDS Math PYQ Chapter ch_1]
(a) 487 (b) 483 (c) 481 (d) 480 85. What

**97.** If A+B= equal to ? x2 —4x+4 2x2 -7X+3 2x-1 2x2+7x-3 2x-1 then what is B 7 the 7 [CDS Math PYQ Chapter ch_1]
(a) 30 (b) 45 (c) 60 (d) 75 23. What IS value of N?

**98.** Ihe product of digits of the number is (c) 10 (d) 12 [CDS Math PYQ Chapter ch_1]
(a) 100066-1 (b) 10000m
(c) 100+-6 (d) 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

**99.** Consider the following statements : 1. sin660 is less than cos660. 2„ sin260 is less than cos260. Which of the above statements is/are corect ? [CDS Math PYQ Chapter ch_1]
(a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2

**100.** If a ^ - l = 5, then what is the value of a^ - 1 [CDS Math PYQ Chapter ch_1]
(a) 125 (b) -125 (c) 140 (d) -140

**101.** The smallest integer with 4 digits which is a perfect square is [CDS Math PYQ Chapter ch_1]
(a) 1000 (b) 1024 (c) 1089 (d) None of the above

**102.** If the average of 9 consecutive positive integers is 55, then what is the largest integer ? [CDS Math PYQ Chapter ch_1]
(a) 57 (b) 58 (c) 59 (d) 60

**103.** 413283P759387, 13 t, 2. 21000000 7 OTT? XVWS-T-MTK/50A 4. 5. 6. 50 100 ('L n) 2 5 (al +a2 + + b2 + b3 + b4) [CDS Math PYQ Chapter ch_1]
(a) 15 (b) 30
(c) 45 (d) 60 2727 -1527*6@avråa OTT? 2

**104.** If LBAC— then what is sine + cos9 equal to ? 71 65 73 65 79 (d) 65 [CDS Math PYQ Chapter ch_1]
(a) I and 11 only (b) 11 and Ill only
(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

**105.** A cricketer has a certain average Of IO innings. In the eleventh inning he scored 108 runs, thereby increasing his average by 6 runs. What is his new average ? [CDS Math PYQ Chapter ch_1]
(a) 42 (b) 47 (c) 48 (d) 60

**106.** A hollow spherical shell is made of a metal of density 7 g/cm3. If its internal and extemal radii are 3 cm and 6 cm respectively, then what is the mass of the shell ? (take [CDS Math PYQ Chapter ch_1]
(a) 2772 g (b) 3322 g (c) 4433 g (d) 5544 g 19 19

**107.** There are two categories of employees in 2019 whose total strength remained the same in 2020. What are these two categories ? [CDS Math PYQ Chapter ch_1]
(a) C and B (b) E and A (c) E and F (d) C and D

**108.** Let A = {7, 8, 9, 10, 11, 12} and B = {7, IO, 14, 15}. What is the number of elements in (A —B) and (B —A) respectively ? [CDS (II) 2012 Q10]
(a) 2 and 4 (b) 4 and 2 (c) 2 and 2 (d) 4 and 4

*— SPACE FOR ROUGH WORK —*

**109.** The area included between nth square and (n — l)th square is independent of n 35 Which of the statements given above is/ are correct ? [CDS (II) 2015 Q14]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

## SECTION : ALGEBRA & POLYNOMIALS (1 QUESTIONS)

**110.** If the area of a square is 2401x^4 + 196x^2 + 4, then what is its side length? [CDS (I) 2011 Q11]

(a) 49x^2 + 3x + 2 (b) 49x^2 - 3x + 2 (c) 49x^2 + 2 (d) 59x^2 + 2

## SECTION : HCF, LCM & LOGARITHMS (34 QUESTIONS)

**111.** If the highest common factor of two positive integers is 24 then their least Common multiple cannot [CDS (II) 2010 Q65]

(a) 72 (b) 216 (c) 372 (d) 600

**112.** The number of digits in 330 is n and it is given that log103 -04771. What is the value of n ? [CDS (II) 2010 Q96]

(a) 13 (b) 14 (c) 15 (d) 16 - A

**113.** There are two numbers which are greater than 21 and their LCM and HCF are 3003 and 21 respectively. What is the sum of these numbers ? [CDS (I) 2011 Q96]

(a) 504 (b) 508 (c) 514 (d) 528

**114.** A can do 50% more work than B in the same time. B alone can do a piece of work in 30 hours. B starts working and had already worked for 12 hours when A joins him. How many hours should B and A work together to complete the remaining work ? [CDS (II) 2013 Q19]

(a) 6 hours (b) 12 hours (c) hours (d) 7-2 hours

**115.** men are supposed to complete a work in 10 days. Aner working for 5 days, they realise that only one-fourth of the work is done. How many more men they need to employ to finish the work on time ? [CDS (II) 2014 Q17]

(a) 40 (b) 30 (c) 20 (d) 15

**116.** What is the greatest possible speed at which a person can walk 13-3 km and 209 km so that the time (in hours) required in each case is a whole number ? [CDS (II) 2014 Q22]

(a) 1-7 km/h (b) 1-9 km/h (c) km/h (d) 2•3 km/h

**117.** From a rectangular metal sheet of sides 25 cm and 20 em, a circular sheet as large as possible is cut-off. What is the area of the remaining sheet ? [CDS (I) 2018 Q17]

(a) 186 (b) 144 (c) 93 cm2 (d) 72 cm2

**118.** ABCD is a square. If the sides AB and CD are increased by 30%, sides BC and AD are increased by 20%, then the area of the resulting rectangle exceeds the area of the square by (b) 52% (c) 540/0 (d) 56% [CDS (I) 2019 Q18]

(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

**119.** Ihe present age of a father is equal to sum of the ages of his 4 children. Aner ten years the sum of the ages of the children will be 1-6 times the age of their father. What is the present age of father ? [CDS (II) 2025 Q19]

(a) 36 years (b) 40 years (c) 42 years (d) 45 years

**120.** If a = xyp—l, b c aq ¯ ¯ P cP — q is equal to = zxr —1, then [CDS (II) 2010 Q21]

(a) ab 1 (b) abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to

(c) abc (d) xyz

**121.** What is the numb' in 125, 823 md 920 ? [Givn hg102 = 0-301, log. —&477, 10867 - 0845] [CDS (II) 2010 Q72]

(a) 21, 20, 19 (b) 20, 19, 18

(c) 22, 21,20 (d) n, 20,21 LA x be integer such that 14 divid« x, is 7; wha 15 x, is

**122.** Wych me Of xver LCM ? [CDS (II) 2010 Q81]

(a) be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ? (b) 80

(c) 60 36 • (d) 24 14. WEch Of

**123.** What is the product of the two numbers ? [CDS (II) 2012 Q16]

(a) 360 (b) 480

(c) 486 (d) 512 Consider the following statements in respect Of an equilateral triangle ABC.

**124.** At the end of the time he received 11,400. How many days did he work ? [CDS (II) 2014 Q20]

(a) 20 (b) 21 (c) 24 (d) 25

**125.** What is value of bgIdsin9) + log10(tnØ) + log I o(cote) + loglßw9) + ? (c) 0-5 [CDS (II) 2017 Q87]

(a) Its lia ixuide O (b) Its lies on the triülgle

(c) It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? (d) AB<RX

**126.** The sum of a number and its square is 20. Then the number is [CDS (I) 2018 Q22]

(a) - 5 or 4 (b) 2 or 3 (c) - 5 only (d) 5 or - 4

**127.** A EErson bought a chair and a table for *750. He sold the chair at a gain of 5% and the table at a gain of 20%. He gained 16% on the whole. What is original cost of table ? [CDS (I) 2019 Q22]

(a) *400 (b) *450 (c) V550 (d) e600

*— SPACE FOR ROUGH WORK —*

**128.** Let a two-digit number be k times the sum of its digits. If thc number formed by interchanging the digits is m times the sum of the digits, then the value Of (b) 10-k (c) 11-k [CDS (I) 2014 Q21]

(a) I only (b) 2 only

(c) Both I and 2 (d) Neither I nor 2 710 — 510 is divisible by 10

**129.** HCF ofp, (p-q) is G Select the answer using the code given [CDS (II) 2016 Q66]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**130.** Select the answer using the code given below : [CDS (II) 2020 Q17]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**131.** If ay — bx _ cx — az _ bz ¯ then which of the following is/are correct ? x+y+z_z 2. a+b+c¯c Select the correct answer using the code given below • [CDS (II) 2021 Q17]

(a) I only (b) 2 only (c) Both I and 2 (d) Neither 1 nor 2 9

**132.** If loglox+logIox2 = 2 log,ox+l, then what is the value ofx ? (d) 10 5 5 [CDS Math PYQ Chapter ch_2]

(a) only (b) k>2v5 only

(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I—aßy

**133.** If (x— 5) is the HCF of —x —p and x2 — qx— IO, then what is the value of (P+q)? [CDS Math PYQ Chapter ch_2]

(a) -23 (b) -17 (c) 17 (d) 23 A -

**134.** A reduction of 10% in the price of sugar enables a person to buy kg more for *2790. What is the reduced price per kilogram ? (b) z48 [CDS Math PYQ Chapter ch_2]

(a) I only (b) 11 only

(c) Both land 11 (d) Neither I nor 11 29. If and then what is log10

**135.** The number Of digits in 330 is n and it is given that log103 -04771. What is the value Of n ? [CDS Math PYQ Chapter ch_2]

(a) 13 (b) 14 (c) 15 (d) 16 - A 10

**136.** There are two numbers which are greater than 21 and their LCM and HCF are 3003 and 21 respectively. What is the sum of these numbers ? [CDS Math PYQ Chapter ch_2]

(a) 504 (b) 508 (c) 514 (d) 528

**137.** What is 8x equal to? 2 4 9 [CDS Math PYQ Chapter ch_2]

(a) Which Of the above statements is/are correct? (b) I only

(c) 2 only (d) Both I and 2

**138.** If tano + = 2, then tang is equal to 3 4 5 4 2 5 (d) 2 - A 16 [CDS Math PYQ Chapter ch_2]

(a) 100066-1 (b) 10000m

(c) 100+-6 (d) 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

**139.** The petpendiculars drawn from any point on OM to AB and AC are always equal Which of the above statements are correct ? [CDS (II) 2009 Q18]

(a) - 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) l, 2 and 3

**140.** If log, 6 m and log, 3 log, (r/2) equal to ? = n, then what is Consider the following statements : A number is divisible by 9 if a, + a2 + + + is divisible by [CDS (II) 2010 Q64]

(a) I only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2 What is x

**141.** HCF(a, b) divides LCM (a, b). Which of the above are correct? [CDS (II) 2010 Q80]

(a) I and 2 Only (b) 3 and 4 only (c) , 2 and 4 only (d) 1, 2, 3 and 4

**142.** What is the HCF of x 3 — 30 and -5x+6? (x +41x-3) (c) (d) [CDS (II) 2010 Q83]

(a) Which Of the above statements is/are correct? (b) I only

(c) 2 only (d) Both I and 2

**143.** The HCFof x —y and xlO—y10 is Which of the statements given above Correct ? are [CDS (II) 2010 Q93]

(a) 1 and 2 (b) 2 and 3 (c) 1 and 4 (d) 2 and 4 26

**144.** If and then what is log10(384) equal to ? (b) p—q+r (d) p+q—r [CDS (I) 2011 Q6]

(a) I only (b) 11 only

(c) Both land 11 (d) Neither I nor 11 29. If and then what is log10

## SECTION : PERCENTAGES, PROFIT, LOSS & DISCOUNT (16 QUESTIONS)

**145.** There are three brothers. The sums of ages of two of them at a time are 4 years, 6 years and 8 years. The age difference between the eldest and the youngest is [CDS (II) 2009 Q25]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

**146.** A 4-digit number N has exactly 15 distinct divisors. What is the total number of distinct divisors of ? [CDS (II) 2009 Q26]

(a) 16 (b) 30 (c) 45 (d) 225

**147.** The minute hand of a clock is 15 cm long and sweeps an area of 15" cm on the dial of the clock. How much angle does it describe during this period ? [CDS (I) 2011 Q23]

(a) 120 (b) 180 (c) 240 (d) 300

**148.** There are two categories of employees whose strength increased by same percentage in 2020. What are the two categories ? [CDS (I) 2012 Q18]

(a) E and F (b) C and E (c) A and E (d) and C A -

*— SPACE FOR ROUGH WORK —*

**149.** In which year was the percentage in- crease in production of cars minimum as compared to its previous year ? [CDS (I) 2012 Q19]
(a) 2015 (b) 2016 (c) 2017 (d) 2018

**150.** What is the percentage of zinc in X ? [CDS (I) 2012 Q21]
(a) 400/0 (b) 36% (c) 34% (d) 30% A -

**151.** Number of credit cards held by an Indlvldual can be treated as [CDS (I) 2012 Q27]
(a) Qualitative data (b) Discrete data (c) Categorical data (d) None of the above

**152.** men working 8 hours a day can completely build a wall of length 100 m, breadth 20 cm and height 5 m in 10 days. How many days will 16 men working 10 hours a day require to build a wall of length 200 m, breadth 60 cm and height 5 m ? [CDS (II) 2015 Q24]
(a) 18 (b) 24 (c) 27 (d) 36

**153.** What is the sum of the radii of the circles ? [CDS (II) 2024 Q29]
(a) 104 cm (b) 11-2 cm (c) 12 cm (d) 13 cm

**154.** A person's salary has increased from e 7,200 toe 8,100. What is the percentage increase in his salary? 23. •Nyo 6 (c) 164% (d) 124% [CDS (I) 2012 Q15]
(a) O only (b) 20 only (c) both 10 and 20 (d) neither IO nor 20 If

**155.** For value of k x3 + — + 6 be C [CDS (I) 2012 Q29]
(a) be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ?
(b) 80
(c) 60 36 •
(d) 24 14. WEch Of

**156.** The price of an article is Rs. 25. After two successive cuts by the same percentage, the price becomes Rs. 20.25. If each time the cut was x%, then (b) x = 10 (c) x = 11 (d) x = 11.5 26. If 0.764 y — 1.236 x, then what is the value of [CDS (II) 2012 Q25]
(a) 0.764
(b) x = 10
(c) x = 11
(d) x = 11.5 26. If 0.764 y — 1.236 x, then what is the value of

**157.** : qvi a > b > O a +b2 a+b a3+b3 a2+b2 2. a2+b2 28. z:nq (3n-1) (211+1) fri x t (3"1+1) (4n-3) 11 t, 29. Q bz—cy a X+y+z z 2. a+b+c¯c NYRG-T-MTK - A 375m2 65 m 3åq [CDS (II) 2014 Q24]
(a) 80 m
(b) 84 m
(c) 90 m
(d) 100m 31. ax2-4ar+15= o 15 2 2 2 21 2 32. ft-3iGrzv 8 1 63 30 km/hrt I

**158.** The difference between their interior angles is [CDS (II) 2023 Q30]
(a) ab 1
(b) abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to
(c) abc
(d) xyz

**159.** If A is odd, then B is odd Select the answer using the code given below : [CDS (II) 2021 Q23]
(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 1 730 1 65

**160.** A person sold an article for e 3,600 and got a profit of 20%. Had he sold the article for 3,150, how much profit would he have got? [CDS Math PYQ Chapter ch_3]
(a) 15
(b) 25
(c) 20
(d) 30 9. 10. 11. What is equal to?

## SECTION : SIMPLE & COMPOUND INTEREST, RATIO & MIXTURES (28 QUESTIONS)

**161.** Which one of the following numbers is not a square of any natural ? [CDS (II) 2009 Q38]
(a) 5041 (b) 9852 (c) 1936 (d) 6241 T-DETA.K.FN - A

**162.** If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the sarru in 2 days, what is the time taken by 15 men and 20 boys in doing the same type of work ? [CDS (II) 2011 Q36]
(a) 4 days (b) 5 days (c) 6 days (d) 7 days

**163.** There are two employees X and Y. X's salary is first increased by 12% and then decreased by 10%, and Y's salary is first increased by 10% and then decreased by 12%. If their salaries at present are equal, then what was the ratio of initial salary of X to initial salary of Y? [CDS (II) 2012 Q96]
(a) 50: (b) 51: (c) 121: (d) 121: 125 I P.T.O.

**164.** The wages, of labourers in a factory has increased in the ratio 22 : 25 and their number decreased in the ratio 3 : [CDS (II) 2012 Q97]
(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

**165.** 10% annum on compound interest for 3 years, compounded annually, and paid back in 3 equal annual installments during these years. What is the amount of each installment (approximately) ? [CDS (I) 2013 Q1]
(a) e 4,437 (b) e 4,237 (c) r 4,021 (d) r 3,811

**166.** In what duration of time will *3300 become *3399 at 12% per annum interest compounded quarterly ? [CDS (I) 2013 Q2]
(a) 3 months (b) 6 months (c) 9 months (d) 1 year

**167.** TWO men, and B run a 4 km racc on a course km round. If their speeds are in the ratio 5 : 4, how often does the winner pass the other ? [CDS (II) 2013 Q89]
(a) Once (b) Twice (c) Thrice (d) Four times - A

**168.** Area of APDA is equal to that of APCQ [CDS (II) 2014 Q37]
(a) — AB (b) a AB2 (c) 1, 2,3 82. 83. (d) 4

*— SPACE FOR ROUGH WORK —*

**169.** cm What is the approximate area of the shaded region in the figure given ? [CDS (II) 2014 Q38]
(a) 15•3 cm2 (b) 25•5 cm2 (c) 284 cm2 (d) cm2

**170.** What is the measure of an interior angle of polygon ? [CDS (I) 2015 Q33]
(a) 1100 (b) 1200 (c) 1300 (d) 1400

**171.** Sex ratio is defined as the number of females per 1000 males. In a place the total inhabitants are out of which 935,000 are females. What is the sex ratio for the place ? [CDS (II) 2016 Q73]
(a) 935 (b) 1000 (c) 1935 (d) 9350 6 2 23

**172.** The length and breadth of a rectangle are in the ratio 4 : [CDS (II) 2016 Q76]
(a) — •E (b) litre (c) litre , (d) — litre

**173.** What is the maximum area of a rectangle, the IVimeter of which is 18cm? [CDS (II) 2017 Q34]
(a) 2025 cm2 (b) 20-00 cm2 (c) 19-75 cm2 (d) 19-60 cm2

**174.** In a pi-diagram there are three sectors. If the ratio of the angles of the sectors is 1 : 2 : 3, then what is the angle of the largest sector ? [CDS (II) 2017 Q45]
(a) 2000 (b) 1800 (c) 1500 (d) 1200

**175.** A tank can be filled by pipe X in 2 hours and pipe Y in 6 hours. At 10 a.m. pipe X was opened. At what time will the tank be filled if pipe Y is opened at 11 a.m. ? [CDS (I) 2019 Q37]
(a) 12:45 hours (b) 5:00 p.m. (c) 11:45 a.m. (d) 11:50 a.m.

**176.** What is c2 + bc—a2 —ab equal to? [CDS (II) 2009 Q34]
(a) 12 21. What is c2 + bc—a2 —ab equal to? (b) -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 —4x— IO equal to? If equal to ?
(c) 2p (d) then what is then what is x 22. If ab ba , then what is equal to?

**177.** IO. 11. 12. Consider the following statements l. Every integer is a rational number. 2. Every rational number is a real number. Which of the above statements is/are correct ? [CDS (I) 2011 Q13]
(a) I only (b) 2 only
(c) Both I and 2 (d) Neither I nor 2 The product of HCF and LCM of 18 and 15 is :

**178.** Which orz of åe mllowing is ca•ræt in Of a light riangle ? [CDS (I) 2012 Q32]
(a) Its lia ixuide O (b) Its lies on the triülgle
(c) It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? (d) AB<RX

**179.** Two cyli&s of have in ratio 2 : [CDS (II) 2012 Q99]
(a) 32% (b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?
(c) cm2 (d) 50

**180.** An alloy A contains two elements, copper and tin in the ratio of 2 : 3, whereas an alloy B contains the same elements in the ratio of 3 : 4. If 20 kg of alloy A, 28 kg of alloy B and some more pure copper are mixed to form a third alloy C which now contains copper and tin in the ratio of 6 ; 7, then what is the quantity of pure copper mixed in the alloy C ? [CDS (I) 2013 Q89]
(a) 3 kg (b) 4 kg (c) 5 kg (d) 7 kg

**181.** The ratio of two complementary angles is I : [CDS (I) 2022 Q58]
(a) 64 minutes (b) 80 minutes
(c) 96 minutes (d) 120 minutes The pair of numbers which are relatively prime to each other is :

**182.** If 63 is added to this number, the digits interchange their places. What is the sum of the digits in the number ? [CDS (I) 2010 Q33]
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2 only when Ware correct ? 28. Let work done by

**183.** Consider the following inequalities : a2—b2 a—b > — where a > b > O a2+b2 a a3+b3 a2+b2 only when 2. a2+b2 a+b Which of the above is/are correct ? [CDS (II) 2010 Q37]
(a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

**184.** If x is a negative real number, then which of the following are not correct ? l. There is some natural number k such that > 0 2. + x > O always 4. x2 is always a rational number Select the correct answer using the code given below : [CDS (II) 2016 Q55]
(a) 1, 2 and 3 (b) 1, 2 and 4 (c) and 4 (d) 2, 3 and 4

**185.** What is the digit in the unit place of the number 11 x 33 x 55 x 77 x 99 ? 13 [CDS Math PYQ Chapter ch_4]
(a) I only (b) 11 only
(c) Both land 11 (d) Neither I nor 11 29. If and then what is log10

**186.** What is p2 + q2 equal to ? (d) 13 (c) 10 43 [CDS Math PYQ Chapter ch_4]
(a) 96. What is cot2ß equal to ? p For the next two (b) 10
(c) 13 For the next two (d) 2pq — —

**187.** cm What is the approximate area of the shaded region in the figure given ? 35 [CDS Math PYQ Chapter ch_4]
(a) 15•3 cm2 (b) 25•5 cm2 (c) 284 cm2 (d) cm2

**188.** A gentleman left a sum of 39,000 to be distributed after his death among his widow, five sons and four daughters. If each son receives 3 times as much as a daughter receives, and each daughter receives twice as much as their mother receives, then what is the widow's share ? [CDS Math PYQ Chapter ch_4]
(a) ^ 1,000 (b) ^ 1,200 (c) 1,500 (d) None of the above

*— SPACE FOR ROUGH WORK —*

## SECTION : TIME, SPEED & DISTANCE, TIME & WORK (21 QUESTIONS)

**189.** A river 2-5m deep and 45m wide is flowing at the speed of km/hour. The amount of water that runs into the sea per minute is [CDS (II) 2013 Q92]

(a) 6650 (b) 6750 (c) 6850 (d) 6950 - A

**190.** A person can row downstream 20 km in 2 hours and upstream 4 km in 2 hours. What is the speed of the current ? [CDS (II) 2013 Q93]

(a) 2 km/hour (b) 2-5 km,'hour (c) 3 km/hOur (d) 4 knvhour

**191.** A motorist travels to a place 150 km away at an average speed of 50 km/hour and returns at 30 km/hour. What is the average speed for the whole journey ? [CDS (II) 2013 Q94]

(a) 35 km,fhour (b) 37 km,fhour (c) 37-5 (d) 40 km/hour

**192.** A car travels from A to B at a speed of 40 km/hr, travels back from B to A at a speed of 30 km/hr and again goes from A to B at a speed of 60 km/hr. What is the average speed of the car? [CDS (II) 2013 Q97]

(a) km/hr (b) 42 km/hr (c) 40 km/hr (d) km/hr

**193.** A train takes two hours less for a journey of 300 km if its speed is increased by 5 knvhr from its usual speed. What is its usual speed ? [CDS (I) 2014 Q31]

(a) 50 knvhr (b) 40 km/hr (c) 35 knvhr (d) 25 km/hr

**194.** There are some benches in a class room having the number of rows 4 more than the number of columns. If each bench is seated with 5 students, there are two seats vacant in a class of 158 students. The number of rows is (d) [CDS (II) 2014 Q45]

(a) ac I If f (b) -20 (c) 20 (d) 12

**195.** By increasing the speed of his car by 15 km/hour, a person covers 300 km distance by taking an hour less than before. The original speed of the car [CDS (II) 2016 Q80]

(a) 45 km/hour (b) 50 km/hour (c) 60 kni/hour (d) 75 km/hour

**196.** A man walking at 5 km/hour noticed that a 225 m long train coming in the opposite direction crossed him in 9 seconds. The speed of the train is [CDS (II) 2016 Q83]

(a) 75 km/hour (b) 80 km / hour (c) 85 km / hour (d) 90 kin/ hour w/ 54A

**197.** When the speed 0fa train is increased by 20%, it takes 20 minutes less to cover the same distance. What is the time taken to cover the same distance with the original speed ? [CDS (II) 2016 Q86]

(a) 140 minutes (b) 120 minutes (c) 100 minutes (d) 80 minutes

**198.** A person rode one-third of a journey at 60 km/hr, one-third at 50 km/hr and the rest at 40 km/hr. Had the person ridden half of the journey at krnfhr and the rest at 40 km/hr, he would have taken 4 minutes longer to complete the journey. What distance did the person ride ? [CDS (II) 2016 Q90]

(a) 180 km (b) 210 km (c) 240 km (d) 300 km

**199.** Without stoppages, a train travels at an average speed of 60 km/h and with stoppages it covers the same distance at an average speed of 48 km/h. HOW many minutes per hour does the train stop ? [CDS (II) 2016 Q95]

(a) 10 minutes (b) 12 minutes (c) 1 S minutes (d) 20 minutes

**200.** A train moving with a speed of 60 km per hour crosses an electric pole in 30 seconds. What is the length of the train in metres ? [CDS (I) 2018 Q27]

(a) 300 (b) 400 (c) 500 (d) 600

**201.** 31-5 km/h 365 km/h 38-5 krn/h 40-5 km/h c In the figure given above, the side of square ABCD is 7 cm. What is the area of the shaded portion, formed by the arcs of the circles with centre at C and A ? [CDS (II) 2023 Q82]

(a) 7 crn2 (b) 28 cm2 (c) 14 cm2 (d) 21 Crn2 - A

**202.** What is the sum Of their reciprocals ? [CDS (I) 2012 Q44]

(a) 64 minutes (b) 80 minutes

(c) 96 minutes (d) 120 minutes The pair of numbers which are relatively prime to each other is :

**203.** If ab ba , then what is equal to? (d) ab 9 n. If x-2+21/2 +23/2 x2 —4x— IO equal to? [CDS (II) 2012 Q43]

(a) 12 21. What is c2 + bc—a2 —ab equal to? (b) -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 —4x— IO equal to? If equal to ?

(c) 2p (d) then what is then what is x 22. If ab ba , then what is equal to?

**204.** If equal to ? 2p 2p then what is then what is x a—b 2 a+b x + b O, then what [CDS (II) 2012 Q44]

(a) 12 21. What is c2 + bc—a2 —ab equal to? (b) -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 —4x— IO equal to? If equal to ?

(c) 2p (d) then what is then what is x 22. If ab ba , then what is equal to?

**205.** What is the of centres of circles which touch a given line at a given point ? [CDS (I) 2025 Q44]

(a) A line pen*ndicular to thc given line, passing through the given (b) A line parallel to the given line

(c) A circle tangent to the given line at the given point (d) A closed curve other -than a circle

**206.** Two students A and B solve an equation Of the form ± px + q = O. A starts with a %T0ng value of p and obtains the roots as 2 and [CDS (I) 2015 Q44]

(a) p (b) 3 and —4 (c) —3 and—4 (d) —3 and 4

**207.** A passenger train and a goods train are running in the same direction on parallel railway tracks. If the passenger train now takes three times as long to pass the goods train, as when they are running in opposite directions, then what is the ratio of the speed of the passenger train to that of the goods train ? (Assume that the trains nm at uniform speeds) [CDS Math PYQ Chapter ch_5]

(a) 2 :1 (b) 3 ; 2 (c) 4 :3 (d) 1 : 1

*— SPACE FOR ROUGH WORK —*

**208.** -Ai if one Of the Of the — bx + c O the square at the then Which one the following is ? 3bc c c Sbe + b [d) 3be 4 b3 + then what is x What ig the value Of O ne Ot factors tf — 5x2 — 81 ia ta) equgi to ? Let A : % is odd integer) and B — : xa — 8x + % — [CDS Math PYQ Chapter ch_5]

(a) 5% (b) 8% (c) 10% (d) 12.5%

**209.** A train moving with a speed of 60 km per hour crosses an electric pole in 30 seconds. What is the length of the train in metres ? [CDS Math PYQ Chapter ch_5]

(a) 300 (b) 400 (c) 500 (d) 600

## SECTION : ALGEBRA, POLYNOMIALS & LINEAR EQUATIONS (6 QUESTIONS)

**210.** Which one of the following is a zero of the polynomial 3x^ + 4x^ - 7 ? [CDS (II) 2015 Q10]

(a) 0 (b) 1 (c) 2 (d) -

**211.** What is the number of students in the class ? [CDS (II) 2024 Q53]

(a) 100 (b) 105 (c) 110 (d) 120

**212.** BC I ABDC = AC.BD, ZBAD = a ZCAD = (c) 2a (d) 28. N- 12345678AB 10-3i*rq t, qvi I 11. t, 30. Q Ilx+5yRE6 (prime number) qvi x, y .f (natural number) (x + y) N 15 [CDS (II) 2015 Q57]

(a) 16 (b) 30

(c) 2a (d) 28. N- 12345678AB 10-3i*rq t, qvi I 11. t, 30. Q Ilx+5yRE6

**213.** If x varies directly as y, then which of the following is/ are correct ? I. (x2 + y2) varies directly as ya. varies directly as y. Select the answer using the code given below : [CDS (I) 2017 Q56]

(a) I only (b) 11 only (c) Both land 11 (d) Neither I nor 11

**214.** If x4 + y4 14x2 , then consider the I. loglO(x 2 + ) loglo x *loglo y t 210g10 2 [CDS Math PYQ Chapter ch_6]

(a) 11 only (b) Both I and 11

(c) Neither I nor 11 39. Which Of the is/ are the Of (d) Both I and 11

**215.** Which one of the following is a zero of the polynomial 3x^ + 4x^ - 7 ? [CDS Math PYQ Chapter ch_6]

(a) 0 (b) 1 (c) 2 (d) - 1

## SECTION : QUADRATIC EQUATIONS & PROGRESSIONS (8 QUESTIONS)

**216.** If a and are the roots of the equation then what is equal to? [CDS (I) 2016 Q18]

(a) 16 (b) 25 (c) 36 (d) 49 1 P.T.O.

**217.** A quadratic polynomial ax^ + bx + c is such that when it is divided by x, (x - 1) and (x + 1), the remainders are 3, 6 and 4 respectively. What is the value of (a + b) ? [CDS (I) 2016 Q70]

(a) 3 (b) 2 (c) 1 (d) -

**218.** B starts with a wrong value Of q and gets the roots as 2 —9. What the correct roots of the equation ? [CDS (I) 2016 Q12]

(a) 3 and — 4 (b) —3 and 4 (c) —3 and 4 (d) 3 and 4 64. of

**219.** If and are the roots of the equation 19, ax2 + bx + c = O, then the value of 1 au + b aß + b bc (b) 20 ab 1 (d) abc Consider the following statements in respect of three a-digit numbers xyz, yzx and ZXY : [CDS (I) 2016 Q14]

(a) ab 1 (b) abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to

(c) abc (d) xyz

**220.** If a and b are positive, then the relation sino 3b (b) Byssible only if a b (c) possible if a > b (d) possible if a < b • 4 [CDS (I) 2019 Q63]

(a) tan29 (b) sin2e COS 9 38. is equal to I — sine tang— I tang +1 1 + sine

(c) coso tano + I tang — I (d) sin 6 39. If tan

**221.** Let p, q be the roots of the equation x2 + mx — n 0 and m, n be the roots of the equation .r2 + px—q = O (m, n, p, g are non-zero numbers). Which of the following statements is/ are correct ? L m(m + n) = IL p+q-l (c) (d) 5 Select the answer using the code given below : [CDS (I) 2017 Q63]

(a) I only (b) 11 only (c) Both land 11 (d) Neither I nor 11

**222.** If X is any point within a square ABCD and on AX a square AXYZ is described, which Of the following is/are corect ? 2 ZABX=ZADZor ZADX= ZABZ Select the correct answer using the code given txlow [CDS (II) 2018 Q60]

(a) I only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2 - A 24

**223.** A quadratic polynomial ax^ + bx + c is such that when it is divided by x, (x - 1) and (x + 1), the remainders are 3, 6 and 4 respectively. What is the value of (a + b) ? [CDS Math PYQ Chapter ch_7]

(a) 3 (b) 2 (c) 1 (d) - 1

## SECTION : TRIGONOMETRY & HEIGHTS AND DISTANCES (13 QUESTIONS)

**224.** ABC is a right triangle with right angle at A. If the value of tan B = any.real k the length of the hypotenuse is of the form (b) 2k (d) 9k (c) 5k [CDS (II) 2017 Q12]

(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

**225.** A pole stands vertically inside a triangular park ABC. If the angle of elevation of the top of the pole from each corner of the park is same, then in the triangle ABC, the foot of the pole is at the [CDS (II) 2017 Q15]

(a) Centroid (b) Circumference (c) Incentre (d) Orthocentre

**226.** sin x cos x 1/2 t, sin x — cos x 80. tan2 y coseca x — 1 = tan2 y t, 81. (d) x —y = 10 [CDS (I) 2017 Q67]

(a) x —y = 10 78. 79. 80. 81. 82. 83. 84. sin x cos x 1/2 t, sin x — cos x DE t (b) *rut A log IO

(c) Tt/2 (d) Tt/3

*— SPACE FOR ROUGH WORK —*

**227.** There is only one Value Of x in the first quadrant that satisfies sin x — cos x = [CDS (I) 2017 Q75]

(a) 64 minutes (b) 80 minutes

(c) 96 minutes (d) 120 minutes The pair of numbers which are relatively prime to each other is :

**228.** Consider the following : cot 300 + 1 = 2 (cos 300+1) cot 300-1 [CDS (I) 2017 Q76]

(a) I only (b) 2 only

(c) Both land2 (d) Neither nor 2 If 3 sin x + 5 cos x = 5, then.what is the value of

**229.** If x=acosO and y=bcot9, then + by-I) is equal to (c) tan29 (d) sin2e COS 9 38. is equal to I — sine tang— I tang +1 1 + sine (b) coso tano + I tang — I (d) sin 6 [CDS (I) 2017 Q84]

(a) tan29 (b) sin2e COS 9 38. is equal to I — sine tang— I tang +1 1 + sine

(c) coso tano + I tang — I (d) sin 6 39. If tan

**230.** What isthevalueof2-!-E cos730 eos71P Sin 170 C - [CDS (I) 2017 Q92]

(a) Its lia ixuide O (b) Its lies on the triülgle

(c) It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? (d) AB<RX

**231.** If o 2' + P) -.13 and tan(a— ± , then what is tana•cot2ß equal to ? [CDS (I) 2017 Q100]

(a) 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e (b) 2h

(c) Htana•tanß (d) h cota•cOtß

**232.** If D is the number Of degrees and R is the number Of radians in an angle O, then which one Of the following is correct ? (b) (c) (d) = IBOR 1801) c 100 1 •50 200 50 20 What is the minimum value Of 9 tan2 + 4 cot2 9 ? It is known that the average price per pencil is 1•25. M'llat is the value Of x ? [CDS (I) 2018 Q50]

(a) 10 (b) e 30 (c) e 40 (d) 60 (a) (c) (d) 6 9 12 13

**233.** For what value Of k is (x + I) a factor of (c) _2 Consider the following Statements : I. There exists a positive real number m such that cos x z 2m 2. mn m + n for all m, n belonging to set of natural numbers. Which of the above statements is/are correct? [CDS (II) 2016 Q41]

(a) I only (b) 2 only (c) Both I and 2 (d) Neither I nor 2

**234.** COS < COS Which Of the above Stater-nents is / are correct? [CDS (I) 2017 Q82]

(a) I Only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2

**235.** For O < —, consider the following : I. (tan40 + + c0160) = sec29 cosec29 tan 0 + sin 0 = cor29(sec9 + 11. tan — sin O Which of the above is/are identities ? [CDS (II) 2017 Q7]

(a) I only (b) 11 only (c) Both land 11 (d) Neither I nor 11

**236.** sin cos (9-20) 6 0 — cos e — cos I — sin2 B cos2 e 2-5 kmph 3aqå 3-25 kmph t I 4 36 [CDS Math PYQ Chapter ch_8]

(a) 16 cm (b) 20 em

(c) 25 cm (d) 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ?

## SECTION : GEOMETRY: LINES, ANGLES, TRIANGLES & POLYGONS (13 QUESTIONS)

**237.** ABC is a triangle right-angled at B. Points M and N trisect BC. If AM = 6 cm and AN = 9 cm, then what is AC equal to ? [CDS (II) 2017 Q21]

(a) 4xfiö cm (b) 265 cm (c) 24 cm (d) 20 cm

**238.** In an equilateral triangle another equilateral triangle is drawn inside joining the mid-points of the sides of given equilateral triangle and the process is continued up to 7 times. What is the ratio of area of fourth triangle to that of seventh triangle ? [CDS (I) 2022 Q77]

(a) 256 : I (b) 128 : (c) 64: (d) 16:1

**239.** What is the area of a triangle with sides of length 12 cm, 13 cm and 5 cm ? [CDS (I) 2022 Q79]

(a) 30 crn2 (b) 35 cm2 (c) 40 cm2 (d) 42

**240.** If the angles of a triangle are 300 and 450 and the included side is + 1) cm, then what is the area of the triangle ? [CDS (I) 2022 Q81]

(a) +1) cm2 (b) + 3) cm2 (c) l) cm2 (d) 266+1) cm2

**241.** The lengths of sides of a triangle are where If one of the angles is 900, then what are the minimum integral values of x, y, z respectively ? [CDS (I) 2022 Q83]

(a) — AB (b) a AB2 (c) 1, 2,3 82. 83. (d) 4

**242.** In , a right triangle ABC, BD is perpendicular on hypotenuse AC If AC 9 cm and AD 4 cm, then what is AB+ BC approximately equal to? [CDS (I) 2022 Q87]

(a) 12 cm (b) 12•2 cm (c) 12•4 cm (d) 126 cm

**243.** The base of an isosceles triangle is 300 units and each of its equal sides is 170 units. Then the area of the triangle is [CDS (I) 2022 Q100]

(a) 9600 square units (b) 10000 square units (c) 12000 square units (d) None of the above

**244.** ABC is a triangle. Let D, E denote the mid points Of BC, CA respectively. Let AD and BE intersect at G Let O be a point on AD such that AO : OD 2 : [CDS (II) 2017 Q10]

(a) Both A and R are individually true and R is the correct explanation of A (b) Both A and R are individually true but R is not the correct explanation of A

(c) A is true but R is false (d) A is false but R is true 98. Assertion

**245.** ABC is a tingle riÖt at B. If ABS 5 cm BC = What is of the p*iølar drawn from B the hypotenuse ? 243 (e) [CDS (I) 2022 Q80]

(a) 32% (b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?

(c) cm2 (d) 50

*— SPACE FOR ROUGH WORK —*

**246.** ABC is a triangle in which AB = AC and D is any point on BC. Which of the following statements is/ are correct ? L AB2-AD2=CDXBD IL BC2+ Select the answer using the code given [CDS (II) 2017 Q20]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**247.** ABC is a triangle right angled at B. P is the midpoint Of AB and Q is the mid- point of BC. Consider the following : I. AQ units 11. units Which of the atxwe is/ are required to determine the area Of the triangle? [CDS (I) 2022 Q93]

(a) I only (b) 11 only (c) Both I and 11 (d) More information is needed

**248.** The quadrilateral AAY is a The area Of the quadrilateral AZXY is half Of the area Of the triangle Which Of the statements given above is/ are correct? [CDS (I) 2022 Q94]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

**249.** Area of triangle OAC is equal to 102 cm2. Select the answer using the code given [CDS Math PYQ Chapter ch_9]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11

## SECTION : OFFICIAL QUESTION PAPER (438 QUESTIONS)

**250.** If each of n numbers x, — i (i = l, 2, 3, ...m) is replaced by (i + l)xi, then the new mean IS n(n + l) 2 (d) [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q1]

(a) 24 (b) —25 (c) 28 (d) —

**251.** For two observations, the sum is S and product is P. What is the harmonic mean Of these two observations ? (b) S A - [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q3]

(a) , B and C (b) , B and D

(c) , E and F 41. The number of three digit numbers (d) Six

**252.** What is the mean of the values ? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q5]

(a) 38 (b) 40 (c) 42 (d) 44 GI. 62. 63. 64.

**253.** The arithmetic mean of n numbers is M. If the sum of first (n — 1) terms is k, then what is the nth number ? (b) nM — k (c) n(M — k) [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q6]

(a) nM — k (b) n (c) 81 (d) 105

**254.** What is the geometric mean of 3, 9, 27, 81, 243, 729, 2187? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q7]

(a) 81 (b) 105 (c) 144 (d) 243

**255.** If 5 is multiplied to each observation, then what Will be the new arithmetic mean? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q8]

(a) 500 (b) 300 (c) 60 (d) 40

**256.** If 10 is added to each observation then what will be the new arithmetic mean ? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q9]

(a) 50 (b) 55 (c) 60 (d) 65 15 A -

**257.** What is the arithmetic mean ? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q10]

(a) 25 (b) 50 (c) 75 (d) 25

**258.** Which one of the following is a factor of the [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q11]

(a) 29, 17, (b) 31, 17, (c) 19, 37, (d) 23, 31,

**259.** A tin of oil was — full. When 6 bottles of oil were taken out from this tin and 4 bottles of oil were poured into it, it was — fill]. Oil of. how many EX)ttles can the tin contain ? (All bottles are of equal volurne) [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q13]

(a) 35 (b) 40 (c) 45 (d) 50

**260.** What is tlE median of the ages Of minor children ? (b) 5 years (c) 7 years (d) Cannot determined due to in- sufficient data [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q14]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

**261.** If a variable takes discrete values a + 4, a-3•5, a-2•5, a-3, a-2, a+O•5, a +5 and a —08 where a > O, then the median of the data set is [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q15]

(a) 24 (b) —25 (c) 28 (d) —

**262.** What is the median of the distribu- tion ? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q17]

(a) 9 (b) 5 (c) 9 (d) 3

**263.** What is the mode of the distribution ? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q18]

(a) I and 11 only (b) 11 and Ill only

(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

**264.** The numbers 59, 53, 51, 43, 36, (8x+ l), (x2+ l), 13, 12, 9, 7, 3 are written in descending order and their median is [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q19]

(a) 25 (b) 50 (c) 75 (d) 25

**265.** If each number is first multiplied by 2 and then increased by 5, then what is the mean Of new numbers ? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q20]

(a) 20 (b) 25

(c) 40 (d) 45 Consider the following types of data

**266.** What is the mean of the ages of minor children ? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q22]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

*— SPACE FOR ROUGH WORK —*

**267.** X is as given below : x frequency [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q24]
(a) 36 (b) 44 (c) 52 64. (d) 150

**268.** The frequency distribution of a variate X is as given below : [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q25]
(a) 36 (b) 44 (c) 52 64. (d) 150

**269.** If 15 is added to each of the first forty values and 5 is subtracted from each of the rernaining sixty values, the new mean becomes : [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q26]
(a) 45 (b) 48 (c) 51 (d) 55

**270.** Consider the following distribution Value of the variable 1 2 3 4 5 naaaa Frequency For What value Of f, is the arithmetic mean of the above distribution 3.1 ? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q27]
(a) Mean Median (b) Mean Mode
(c) Mean > Median (d) Mean > Mode What is the harmonic mean 25, 40 and 50 ?

**271.** A distribution consists of 3 components With frequencies 45, 40 and 55 having their means 2, 2 • 5 and 2 respectively. What is the mean of the combined dis tribu tion? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q36]
(a) 2 (b) 225 (c) 2 (d) 2

**272.** The quantity which must be added to (1— x 2) to obtain x 3 is 2X3 + 3X2 + I 2x3+x + x —1 (b) Consider the following statements • [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q42]
(a) 15 (b) 25 (c) 20 (d) 30 9. 10.

**273.** Consider •the following frequency distributién • Class 0-10 0-20 0-30 0-40 0-50 Frequency 3 What is the above frequency distributioÖ known as? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q52]
(a) umulative distribution in more than type (b) umulåtive distribution in less than type
(c) ontinuous frequency distri- bu tion (d) None Of above / 26A

**274.** Which of the following is/are correctly matched ? [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q70]
(a) Mean Median (b) Mean Mode
(c) Mean > Median (d) Mean > Mode What is the harmonic mean 25, 40 and 50 ?

**275.** If A+B= equal to ? x2 —4x+4 2x2 -7X+3 2x-1 2x2+7x-3 2x-1 then what is B 7 the [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q71]
(a) 30 (b) 45 (c) 60 (d) 75

**276.** The weighted arithmetic mean of first 10 natural numbers whose weights are equal to the corresponding numbers is equal to (d) 385 7 [UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper Qcds_2026_ii_math_q88]
(a) 24 (b) —25 (c) 28 (d) —

**277.** If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? (Take It = —I [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q2]
(a) 242 cm (b) 385 cm (c) 462 cm (d) 770 cm

**278.** cubes each of volume 343 cm3 are joined end to end. What is the total surface area of the resulting cuboid ? [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q3]
(a) 343 cm2 (b) 350 cm2 (c) 686 cm2 (d) 700 cm2

**279.** A tub is in the sharx of a frustum Of a cone. The radii of two circular ends of the tub are 105 cm and 42 cm. If the vertical height of the tub is 16 cm, what is its slant height ? [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q4]
(a) 63•5 cm (b) 65 cm (c) 5 cm (d) 75 cm

**280.** Let x, y, z (all prime) be the length, breadth and height respectively of a cuboid with x > y > z. The volume of the cuboid is 30k3 cubic units, where k is a natural number. What is the total surface area of the cuboid ? [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q5]
(a) 60 square units (b) 62 square units
(c) 64 square units (d) nnot be determined due to insufficient data

**281.** If u,.v, w are real numbers such that u3—8v3 18uvw, which one of the following is conect ? (d) u +2v=-3w [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q6]
(a) 3 years (b) 4 years (c) 5 years (d) 6 years

**282.** W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ? [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q7]
(a) 2280 (b) 2640 (c) 3820 (d) 4130

**283.** If the height of a cone•is increased by 50%, then what is ihe percentage increase in the volunie Of the cone? [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q9]
(a) 100/3 (b) 40 (c) 50 (d) 200/3

**284.** If then what•is one of the values of x equal to? [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q10]
(a) 15 (b) 25 (c) 20 (d) 30 9. 10.

**285.** A person saves more than he did the previous year. If he saves *2000 in the first year, in how many years will he save 170000 ? [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q11]
(a) 16 years (b) 17 years (c) 18 years (d) 19 years

**286.** Three cubes each Of side 5 cm are joined end to end. What is the surface area Of the resulting cuboid ? [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q12]
(a) 300 cml (b) 350 cm2 (c) 375 cm2 (d) 400 cm2

*— SPACE FOR ROUGH WORK —*

**287.** Let be a pyramid on a square base and B be a cube. Let a, b, c denote the number of edges, number of faces and number of corners respectively. Then the result a = b + c is true for [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q15]

(a) only (b) only (c) oth A and B (d) neither A nor B

**288.** What is the surface area of the double cone so formed? (c) 1310-4 cm 2 (d) 13188 cm2 [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q16]

(a) The triangle is isosceles (b) The triangle is equilateral

(c) None Of the ab ove For the next three (d) 3124

**289.** Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive y-axis. 'Ihe cone is filled with water up to half of its height. The volume of water is 8 [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q18]

(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

**290.** Ifthesurface area ofa cube is 13254 cm2, then the length of its diagonal is [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q19]

(a) 4445 cm (b) 16 cm (c) 4745 cm (d) 476 cm

**291.** If the total surface area is 616 cm2, what is the volume of the cylinder ? [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q20]

(a) 539 cm3 (b) 616 cm3 (c) 1078 cm3 (d) 1232 cm3

**292.** What is the slant height of the cone ? [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q21]

(a) 60 m (b) 65 m (c) 68 m (d) 70m

**293.** A cinular a Of 20 cm its curved na is 1000 cml. What is ere volume Of [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q22]

(a) 40m cm' (b) 4500 cm3 (c) 5000 cm3 (d) m3

**294.** The number 2x3*5x7x11+1 is [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q30]

(a) prime number (b) not a prime, but power of a prime

(c) not a power of a prime, but a composite even number (d) not a of a prime, but a composite odd number

**295.** If 0.764 y 1.236 x, then what is the Y¯x value Of [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q32]

(a) x- (b) x =

(c) x = 11.5 If 0.764 y = 1.236 x, Y—x value Of (d) 0.764

**296.** If 96-64a3 3 = what is alt + 4a3 equal to ? then [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q34]

(a) 30 (b) 45 (c) 60 (d) 75

**297.** D In the above figure, ABCD is a parallelo- gram. p is a point on BC such that PB : PC=I : [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q55]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

**298.** If BD is the perpendicular on the side AC, then what is the length of BD ? 1008 65 (b) cm 65 A - [UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper Qcds_2026_i_math_q56]

(a) I and 11 only (b) 11 and Ill only

(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

**299.** If the diagonals Of a rhombus are X and y, then what is its area ? 2 (b) 4 (c) xy [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q2]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

**300.** The area of a rhombus is 336 square cm. Ifthe length ofone of its diagonals is 48 cm, then what is the perimeter of rhombus ? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q3]

(a) 200 cm (b) 120 cm (c) IOOcm (d) 90 cm A - 2 sin39— sino

**301.** ABCD is paralle I square metre square metre square metre square metre a trapezium in which AB is to DC Let E and F be the midpoints on AD and BC respectively. If EF IO cm and AB— DC 4 cm, then what is the value of ABX DC? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q4]

(a) 84 square cm (b) 96 square cm (c) 100 square cm (d) 108 square cm

**302.** If a equation xa — x — 1 O, then what is equal to? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q6]

(a) 56 (b) 60 cm (c) 64 cm (d) 1 only

**303.** What are these two categories ? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q8]

(a) nd B (b) E and A (c) E and F (d) nd D

**304.** ABCD is a parallelogram where AC and BD are the diagonals. If ZBAD 600, ZADB 900, then what is BD2 equal to ? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q12]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

**305.** A line through the vertex A of a parallelogram ABCD meets DC in p and BC produced in Q. If P is the mid-point of DC, then which of the following Ware correct ? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q13]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

**306.** Which the Only 2 only Neither Q-DETA.N.FMN Stet-em correct ? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q17]

(a) x = 1, y = (b) x = 2, y = (c) x = 3, y = (d) x = 1, y =

**307.** Between 3 and 4 0'clock, both hour hand and minute hand will coincide past 3 0'clock between [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q20]

(a) 16 minutes (b) 17 minutes (c) 18 minutes (d) 19 minutes

*— SPACE FOR ROUGH WORK —*

**308.** If p divi&s qr if p divides q, p must divi& r. 2' 2' WNch of the [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q27]
(a) 1 only (b) 96" cm2 (c) 2 only (d) oå 1 2 69T cm2

**309.** If a set Acontains 60 elements and another set B contains 70 elements and there are 50 elements in common, then how many elements does A u B contain ? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q28]
(a) 130 (b) 100 (c) 80 (d) 70

**310.** If two factors of — 2a3 — 9a2 + 2a + 8 are (a + l) and (a — l), then what are the Other two factors ? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q29]
(a) — 2) and (a + 4) (b) + 2) and (a + 4) (c) + 2) and (a — 4) (d) — 2) and (a —4)

**311.** If the average marks Of the complete class is 72, then what is the average marks Of the girls ? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q38]
(a) 64 (b) 65 (c) 68 (d) 74

**312.** What is Z BAP equal to ? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q63]
(a) 300 (b) 400 (c) 450 (d) 600

**313.** A person purchases one kg of tea powder from each of the four places A, B, C, D at the rate per 1 kg, 2 kg, 4 kg, 5 kg. If on an average he purchased x kg of tea powder per 1000, then what is the approximate value of x ? [CDS (II) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_ii_math_q66]
(a) 195 (b) 00 (c) 205 (d) 2•10

**314.** What is the least value of n if 194480+nzm4, where m and n are natural numbers? If then What is the value Of ? What is the area Of the region enclosed by three identical circles (each of radius 4 cm) touching each other? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q1]
(a) — square cm (b) 16vTä— — square cm (c) (160 — 8r) square cm (d) — square cm

**315.** A hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 what is the weight of the hemi- sphere(l cm3 of the metal weighs 9 g) ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q2]
(a) 54 g (b) 366 g (c) 122 g (d) 1087t g

**316.** The material Of a solid cone is converted into the shape of a solid cylinder Of equal radius. If the height of the cylinder is 5 cm. what is the height Of the cone ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q3]
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm

**317.** A Wire of diameter 5 mm is wound around it, so as to cover it completely. What is the length of the wire used ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q4]
(a) 22 m (b) 20 m (c) 18 m (d) None Of the above

**318.** If the diameter of a sphere is doubled, how does its surface•area change ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q5]
(a) It increases two times. (b) It increases three times. (c) It increases four times, (d) It increases eight times.

**319.** What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q6]
(a) 256 (b) 512 (c) 768 (d) 1024

**320.** p(x) q(x) (LCM) (HCF) —a3) (x2—ax+a2) I x4+a2x?+a4 t, q(x) [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q7]
(a) x4 —2a2x2 + a4 (b) x4 + 2a2x2+ a4
(c) x2—a2 (d) x4 —a2x2 + a4 x3+2x2-5X+2 x3 + 4x2 + 11. x+l

**321.** The thickness of a cylinder is I foot, the inner radius of the cylinder is 3 feet and height is 7 feet. TO paint the inner surface it requires one litre of a particular colour. How much quantity of the same colour is required to paint all the surfaces of the cylinder ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q8]
(a) litre (b) litre , (c) — litre (d) — litre

**322.** A cone of height 16 cm and diameter 14 cm is mounted on a hemisphere of same diameter. What is the volume of the solid thus formed ? (take [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q9]
(a) 1540 cm3 (b) 1078cm3 (c) 1048 cm3 (d) 770 cm3

**323.** A cubical block of side 14cm is sur- mounted by a hemisphere of radius 7 cm. What is the total surface area of the solid thus formed ? (take 9) [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q10]
(a) 1330 cm2 (b) 1306 cm2 (c) 1296 cm2 (d) 1256 cm2

**324.** How many silver coins, cm in diameter and of thickness 4 mm, must be melted to form a cuboid of dimen- sions 21 cm x 11 cm x 7 cm ? (take 4) [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q11]
(a) 420 (b) 210 (c) 200 (d) 168 A - fiiü 12%

**325.** A sphere of diameter 14 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 14 cm. If the sphere is completely submerged in water, then how much will the level of water rise ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q13]
(a) — cm (b) — cm (c) — cm (d) — cm

**326.** A hemispherical bowl of internal radius 20 cm contains sauce. The sauce is (o be filled in conical shaped bottles of radius 5 cm and height 8 cm. What is the number of bottles required ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q14]
(a) 100 (b) 90 (c) 80 (d) 75

**327.** What is åe of bikers in mad 2017 ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q16]
(a) 163 (b) 152 (c) 141 (d) 134

*— SPACE FOR ROUGH WORK —*

**328.** A circle touches all the four sides of a quadrilateral ABCD. If AB— 9 cm, BC 8 cm and CD — 12 cm, then what is DA equal to? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q18]

(a) 14 cm (b) 13 cm (c) 12 cm (d) I lcm

**329.** ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q19]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

**330.** In which year the difference in their prices exceeded 0000 for the first time ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q20]

(a) 2012 (b) 2013 (c) 2014 (d) 2015

**331.** If the annual income of X is 20% more than that of then the income of Y is less than that of X by pO/o. What is the value of p ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q21]

(a) 10 (b) 16i (c) 36 (d) 27

**332.** A boy is cycling such that the wheels of the Cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, the speed of the cycle is [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q22]

(a) 5 km/hour (b) 15•84 kn/hour (c) 16 km/hour (d) 36 km/hour

**333.** Ihe wheels Of a car are of diameter 80 cm each. The car is travelling at a speed of 66 km,'hour. What is the number of complete revolutions each wheel makes in IO minutes ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q23]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

**334.** A semi circular thin sheet of a of diameter 28 cm is bent and an open conical cup is made. What is the capacity of the cup ? 1000 [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q24]

(a) m3 (b) 3006 cm3 (c) m3 1078 (d) m3 - A

**335.** Colour of the eye Of the person [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q25]

(a) 21 (b) 24

(c) 36 (d) 42 Which one among the following is not correct ?

**336.** The data collected by investigator to be used by himself are called primary data [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q27]

(a) 21 (b) 24

(c) 36 (d) 42 Which one among the following is not correct ?

**337.** Into a conical tent Of radius 8•4m and vertical height m, how many full bags of wheat can be emptied, if space required for the wheat in each bag is [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q29]

(a) 264 (b) 201 (c) 132 (d) 105

**338.** Rain water from ao roof 22 m x 20 m drains into a cylindrical vessel having diameter of base 2 m and height 3-5 m. If the vessel is just full, what is the rainfall ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q30]

(a) Rs. (b) Rs. (c) Rs. (d) Rs.

**339.** HOW many cubic metre Of earth is to be dug out to dig a well of radius 14 m and [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q31]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

**340.** What is the sum of the linear factors (in x and y) of the expression 2x-2 + — 3y2 ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q32]

(a) 2t-3y (b) 3x-2y (c) 3x + 2y (d) 2r+3y

**341.** For which values of k, does the equation x2 — kx + 2 0 have real and distinct solutions ? (b) only (c) k>2v5 only (d) k<-24i or k>2Vi [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q33]

(a) only (b) k>2v5 only

(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I—aßy

**342.** If a + D + Y aß+ "y + ya, then what is (1 equal to? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q34]

(a) only (b) k>2v5 only

(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I—aßy

**343.** A trolley with two wheels one metre apan is moved clockwise on the circular track around a ground with radius 50 m (described by right wheel). If the size of each wheel is of 1 foot radius and the right wheel turns 1000 times, how many times will the other wheel tum ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q35]

(a) 1010 (b) 1015 (c) 1020 (d) 1025

**344.** ABC is a triangle right-angled at C. A semicircle is drawn on AB as diameter. Let P be any point on AC produced such that AP = AB IO cm. Further, B and P are joined. If BC 8 cm, then what is BP equal to ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q36]

(a) 443 cm (b) 9 cm (c) I Ocm (d) m

**345.** In the figure given above, O is the centre Of a circle circumscribing a quadrilateral ABCD. If AB BC and ZBAC 400, then what is Z A DC equal to ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q37]

(a) 500 (b) 600 (c) 700 (d) 800

**346.** A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p, then the area of the polygon is (b) 128 (b) (d) 2 None of the above [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q39]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

*— SPACE FOR ROUGH WORK —*

**347.** What the maximum number of circum-circles that a triangle can have ? (d) Infinite [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q40]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

**348.** The radius of circum-circle of a right- angled triangle is 10cm and the altitude drawn to the hypotenuse is 8 cm. What is the area of the triangle ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q41]

(a) m2 (b) 80 cm2 (c) 100 cm2 (d) 120 cm2

**349.** What is the number of sides of the polygon? 9 (b) 10 (c) 11 (d) 12 ,'45A [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q43]

(a) 15 200 73. (b) 10 (c) 11 (d) 12

**350.** 5 km/h 365 km/h 38-5 krn/h 40-5 km/h c In the figure given above, the side of square ABCD is 7 cm. What is the area of the shaded portion, formed by the arcs of the circles with centre at C and A ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q45]

(a) 7 crn2 (b) 28 cm2 (c) 14 cm2 (d) 21 Crn2 - A

**351.** A wire Of length 36 cm is bent in the form of a semicircle. What is the radius Of the semicircle? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q46]

(a) 9 Crn (b) 8 Crn (c) 7 Crn (d) 6 Ctn

**352.** The chord of an arc of a circle is of length x, the height of the arc is y and the radius of the circle is z. Which one Of the following is correct? y(2z— y) xa y(2Z— y) = 4X2 (b) 2ygz-y1 x2 (c) 4yPz— x2 (d) . In a triangle ABC, ZB-2ZC=2ZA. [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q50]

(a) 2 only above statements (b) /5 (c) /3 (d) /4

**353.** A circle and a square have the same perimeter. Which one of the following is correct ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q51]

(a) Their areas are equal (b) The area Of the circle is larger

(c) The area of the square is — times area of circle (d) The area of the square is It times area of circle

**354.** ABCDA is a con-cyclic quadrilateral of a circle ABCD with radius r and centre at O If AB is the diameter and CD is parallel and half of AB and if the clrcle completes one rotation about the centre O, then the locus of the middle point of CD is a circle of radius 3r 2r - A 20 2-vfir 3 2 , then what is [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q52]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

**355.** In a circle of radius 5 cm, AB and AC are two chords such that AB 8 cm. What is the length of chord BC? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q54]

(a) +1 (b) + (c) l (d) 266+1

**356.** A circle of diameter 8 cm is placed in such a manner that it touches two perpendicular lines. Then another smaller circle is placed in the gap such that it touches the lines and the circle. What is the diameter of the smaller circle ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q55]

(a) — •E (b) litre (c) litre , (d) — litre

**357.** If an arc of a circle of radius 6 cm subtends a central angle measuring 300, then which one of the following is an approximate length of the arc ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q56]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

**358.** What is the area of the shaded region in the given figure, ifthe radius of each Of the circles is 2 cm ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q57]

(a) — cm2 (b) — cm2 (c) — E cm2 (d) m2

**359.** What is the approximate area of minor segment of the circle ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q58]

(a) 2•71 cm2 (b) 2•42 cm2 (c) 1•91 cm2 (d) 1•71cm2 A -

**360.** The area of the circle circumscribing three identical circles touching each other is square cm. What is the radius of one of the smaller circles ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q60]

(a) m (b) I cm (c) m (d) m A -

**361.** A circle is inscribed in a triangle ABC right-angled at B. If AB 5 cm and BC = 12 cm, then what is the radius of the circle ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q61]

(a) 121 (b) 120 (c) 119 (d) None of the above

**362.** What is the value of LAOC ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q63]

(a) 540 (b) 520 (c) 390 (d) 270

**363.** Four equal-sized maximum circular plates are cut off from a square paper sheet Of area 784 square Cm. The circumference of each plate is [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q65]

(a) I Ctn (b) 22 cm

(c) 33 cm (d) 44 W/ 52A 20 58. is equal to — I, then (a)

**364.** What is area of. largest triangle inscribed in a semi-circle Of radius r units ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q67]

(a) square units (b) 2r2 square units (c) 3? square units (d) 4r2 square units

**365.** If area of a circle and a square are same, then What is the ratio Of their perimeters ? 2 4 A - [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q68]

(a) +1 (b) + (c) l (d) 266+1

**366.** What is the area of the largest square plate cut from a circular disk Of radius one unit ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q69]

(a) 4 square units (b) 245 square units (c) r square units (d) 2 square units

*— SPACE FOR ROUGH WORK —*

**367.** A drinking glass Of height 24 cm is in the shape of frustum of a cone and diameters of its bottom and top circular ends are 4 cm and 18 cm respectively. If we take capacity Of the glass as xx cm , then what is the value Of x ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q76]

(a) 824 (b) 1236 (c) 1628 (d) 2472

**368.** The sum of the height and the radius Of a right circular cylinder is 2 1 cm, and the radius is less than the height. If the curved surface area of the cylinder is 616 cm2, then What is the volume Of the cylinder? (Take It 22 / [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q81]

(a) 1078 cm (b) 1617 cm (c) 1927 cm (d) 2156 cma

**369.** Which one of the following statements is correct ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q84]

(a) Mahesh saves more than what Kamal saves (b) Savings of both of them are equal

(c) Kamal saves more than what Mahesh saves (d) It is not possible to determine who saves more

**370.** The average score of class X is [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q85]

(a) 150 (b) 300 (c) 450 (d) 600

**371.** The average score of class Z is [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q86]

(a) 150 (b) 300 (c) 450 (d) 600

**372.** What is the initial rate of interest per annum ? [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q87]

(a) Rs. 1,200 (b) Rs. 1,440 (c) Rs. 1,500 (d) Rs. 1,800

**373.** If the digits are reversed,' the number is increased by [CDS (I) 2025 : Elementary Mathematics Real Test Paper Qcds_2025_i_math_q88]

(a) 100066-1 (b) 10000m (c) 100+-6 (d) 24.16 m

**374.** If its internal and extemal radii are 3 cm and 6 cm respectively, then what is the mass of the shell ? (take [CDS (II) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_ii_math_q3]

(a) 2772 g (b) 3322 g (c) 4433 g (d) 5544 g

**375.** What is AB2 + 4BQ2 equal to ? [CDS (II) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_ii_math_q4]

(a) 41•25cm2 (b) 42•25 cm2 (c) 75 cm2 (d) 44•25 cm2

**376.** What is AQ2 + CP2 equal to ? (b) (c) (d) 1-2 AC2 1-25 AC2 1-5 AC2 A - [CDS (II) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_ii_math_q5]

(a) 10•05 cm2 (b) 10•15cm2

(c) 05 cm2 (d) 11•15cm2 Consider the following for the next three

**377.** The Of two number' is 6 10 12 15 Of N:uation• y O a unique x unique X infinite solutions -2+4 Za-l 2.3—28-1 2.8—'28 + 292 •a + — Qy) only O.OETA [CDS (II) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_ii_math_q60]

(a) k = (b) k = (c) k = (d) k = ±4

**378.** 3bc c c Sbe + b [d) 3be 4 b3 + then what is x What ig the value Of O ne Ot factors tf — 5x2 — 81 ia ta) equgi to ? a unique x a unique solution x — infinite solutions 2.3-28-1 2.8-'28 291 + [CDS (II) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_ii_math_q61]

(a) x = 1, y = (b) x = 2, y = (c) x = 3, y = (d) x = 1, y =

**379.** A hollow cylindrical drum has internal diameter of 30 cm and a height of 1 m. What is the maximum number of cylindrical boxes of diameter 10 cm and height IO cm each that can be packed in the drum ? [CDS (II) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_ii_math_q62]

(a) 60 (b) 70 (c) 80 (d) 90

**380.** If then what is (a + b) : (a — b) equal to ? [CDS (II) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_ii_math_q67]

(a) 3 only (b) 5 only (c) —3 or (d) —5 or

**381.** A hemispherical bowl of intemal radius 18 cm contains a liquid. The liquid is filled in small cylindrical bottles of internal radius 3 cm and internal height 4 cm. What is the number of bottles used to empty the bowl ? [CDS (II) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_ii_math_q69]

(a) 54 (b) 81 (c) 108 (d) 135

**382.** What is the square root of 23 —4•v'fiS ? (b) 7-3-6 [CDS (I) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_i_math_q22]

(a) 29, 17, (b) 31, 17, (c) 19, 37, (d) 23, 31,

**383.** A sum of Rs. 10,000 is deposited for 1 year at the rate of interest 10% com- pounded half yearly. What will be the interest at the end of one year ? [CDS (I) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_i_math_q26]

(a) Rs. 1000 (b) RK1025

(c) Rs. 1050 (d) RS. 1100 a mixture of milk and water of

**384.** Consider the following in respect of the equation y = l. if x > 1 2. if x < 1 3. y exists for all values of x Which of the above statements is/are correct ? [CDS (I) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_i_math_q31]

(a) I only (b) 2 only

(c) I and 2 only I, 2 and Y is/are 14. If (d) 2 minutes

**385.** How much less amount is allocated to A and C together as compared to D and E together? [CDS (I) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_i_math_q39]

(a) 0004 lakhs (b) 0005 lakhs (c) 6008 lakhs (d) 0010 lakhs 45K

**386.** For how many real values of k is 2 + 12kx 16 a "uare for every integer x ? [CDS (I) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_i_math_q87]

(a) Zem (b) One (c) Two (d) Four

**387.** If equal to? 195 16 16 then What is If the equation 4x 2 — 2kx + 3k O has equal roots, then What are the values of k? [CDS (I) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_i_math_q88]

(a) Zem (b) One (c) Two (d) Four

*— SPACE FOR ROUGH WORK —*

**388.** How many 200 mm lengths can be cut from IO m of ribbon? [CDS (I) 2024 : Elementary Mathematics Real Test Paper Qcds_2024_i_math_q96]
(a) 50 (b) 40 (c) 30 (d) 20

**389.** What is the average marks Of all the candidates? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q2]
(a) 758 (b) 8 (c) 760. (d) 768'.

**390.** AD2 — 3CD2 AC2 = AD2 [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q4]
(a) 121-7 121-75 (b) 121-75 ABC t, X Y AX.AYur (c) 16 (d) 20 cm

**391.** In Which one Of the following pairs of years, the difference in total number Of tablets produced between them is m inimum? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q11]
(a) (2003, 2005) (b) (2001, 2005) (c) g003, 2004) (d) (2000, 2002)

**392.** A vertical pole of length 80m is situated on the horizontal plane. The base of the pole is at P. There are two points A and B such that p, A, B are on the same straight line. Let the angles of elevation of top of the pole from A and B be a and (a>ß) respectively. If PA 64 m and AB 36 m, then what is (a +ß) equal to ? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q12]
(a) 600 (b) 900 (c) 1200 (d) 1350 15 A -

**393.** If a and are complementary angles, then which one Of the following is [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q13]
(a) mn — I O For the following two (b) 2/3 For the following tvo
(c) 1 P.T.O. For the following two (d) 2/3 sina 4 u sin # cosa cosß

**394.** The median BD of ghe triangle ABC meets at D. If then which one Of the following is correct ? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q17]
(a) ZACB = I right angle (b) LBAC = I right angle (c) LABC I right angle (d) None of ttw abové - A

**395.** Consider the follOW'ing statements • If two mangles are equi- angular, then they are similar. [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q19]
(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square
(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

**396.** (b) 350 (c) 400 (d) 45 Let ABC be a triangle with AB = 3 Cm, AC = 5 cm. If AD is a median drawn from the vertex A to the side BC, then which one of the following is correct ? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q21]
(a) is always greater than 4 cm but less than 5 cm (b) is always greater than 5 cm
(c) is always less than 4 cm (d) None of the above Consider the following statements

**397.** If every side Of an equilateral triangle is doubled, then the area Of new triangle becomes k times the area Of the Old One. What is k equal to? 17 (d) . The 7 4 3 sides Of a triangle are in the ratio — : If its perimeter is 52 cm, then What is the length Of the smallest side? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q23]
(a) 9 cm (b) I O Crn (c) 11 cm (d) 12 cm

**398.** How many right angled triangles can formed by joining the vertices of a cuboid ? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q24]
(a) 24 (b) 28 (c) 32 (d) None of the above

**399.** What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q29]
(a) 30 crn2 (b) 35 cm2 (c) 40 cm2 (d) 42

**400.** Then what is the ratio of the area of the triangle formed by the parts of the diagonals with a long side to the area of the triangle formed by the parts of diagonals with a short side ? (b) 4.3 (c) 16:9 [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q33]
(a) — •E (b) litre (c) litre , (d) — litre

**401.** ABC is a triangle right-angled at B with AB 8 cm and BC = 6 cm. It is made to revolve about its side BC. What is the approximate total surmce area of the cone so formed ? (take [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q35]
(a) 452 cm2 (b) 440 cm2 (c) 432 cm2 (d) 420 cm2

**402.** What is the length of the altitude of the triangle drawn from vertex A on BC ? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q36]
(a) 10•05 cm2 (b) 10•15cm2
(c) 05 cm2 (d) 11•15cm2 Consider the following for the next three

**403.** If p, q and r are the lengths (in cm) of the sides of a right-angled triangle, then (p — q — — r — — p — q) (d) 16 is always [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q38]
(a) Positive only (b) Negative only (c) Non-positive only (d) Non-negative only

**404.** The base of a right-angled triangle is times the height of triangle. If the area of the triangle is 54 square cm, then what is the perimeter of the triangle ? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q39]
(a) 30 cm (b) 32 cm (c) 36 cm (d) 40 cm

**405.** The ratio of sum of interior angles to sum of exterior angles of a regular polygon of n sides is [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q40]
(a) 1100 (b) 1200 (c) 1300 (d) 1400

**406.** In a right triangle ABC, 900 and AD is perpendicular to BC. If ECAD 600 and BC = 6 cm, then what is AB equal to ? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q41]
(a) 1100 (b) 1200 (c) 1300 (d) 1400

**407.** In a triangle ABG ZA 300, AB 7 cm and AC• 12 Ctn. What is the area of the triangle ABC ? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q45]
(a) 21 cm (b) 210 cm2 (c) 42 cm (d) 420 cm2

*— SPACE FOR ROUGH WORK —*

**408.** ABC is a triangle right angled at B. D is a point on AC such that BD is perpendicular to AC If AB p and BC v'äp, then what is BD equal to? (b) p 12 (d) Op/4 [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q46]

(a) 15 200 73. (b) 10 (c) 11 (d) 12

**409.** ABC is a triangle right-angled at C. Let p be the midpoint of BC. If AP cm and AB 20 cm, then what is the perimeter of the triangle ABC? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q47]

(a) 40 cm (b) 48 cm (c) 50 cm (d) 60 cm

**410.** A square and an equilateral triangle have equal perimeter. If the diagonal of .the square is 12-45 cm, then the area of the triangle is [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q48]

(a) m2 (b) 240 CITI (c) 480 cm2 (d) 6443 cm2

**411.** DP and when produced meet at Q. If area of triangle BPQ is 20 square unit, the area of triangle DC? is [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q49]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

**412.** The base Of an isosceles triangle is 300 units and each of its equal sides is 170 units. Then the area Of the triangle is [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q50]

(a) 9600 square units (b) 10000 square units (c) 12000 square units (d) None of the above

**413.** If the perimeter of an isosceles right triangle is 4 (2 cm, then What is its area in square cm? The 8 16 24 diagonal of a square is 120 cm and the area of an equilateral triangle is 64" Square Cm. Which Of the following • statements is/ are correct? part? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q51]

(a) 150 (b) 225 (c) 300 (d) 450

**414.** What is the area of triangle OMN? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q52]

(a) 36 square cm (b) 40 square cm (c) 45 square cm (d) 48 square cm

**415.** In a triangle ABC, 21 cm, BC—20cm and CA = 13 cm. A per- pendicular CD is drawn upon the longest side. What is the area of the triangle BCD ? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q53]

(a) 96 square cm (b) 84 square cm (c) 80 square cm (d) 72 square cm

**416.** Frequency polygon Frequency curve Frequency distribution All of the above D E Three straight lines X, Y and Z are parallel and the angles are as shown in the figure above. What is ZAFB equal to ? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q54]

(a) 200 (b) 150

(c) 300 (d) 100 - A 24 (a) (b) (c) (d) (a) 1 (b) 2 (c) 3 (d)

**417.** HOW many pairs Of positive integers m and n satisw the equation m n 12 where n is an odd integer less than 60? 17 (d) . The 7 4 3 sides Of a triangle are in the ratio — — : If its perimeter is 234 52 cm, then What is the length Of the smallest side? [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q55]

(a) 9 cm (b) I O Crn (c) 11 cm (d) 12 cm

**418.** If a and b are negative numbers and c i s a pos itive real num , then Which of the following is/ are c [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q57]

(a) M < Median (b) GM Median

(c) M = Me dian Suppose for Which one of the corre ct? (d) GM>1/6

**419.** If is an integer and a, are the roots of 4x2 — such that I < a < 2 4 and 2 < < 3, then how many values can take ? (c) 14 (d) 15 62 +72 +82 +92 +102 [CDS (II) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_ii_math_q58]

(a) x = 100 (b) x = IO (c) 330 (d) 340

**420.** If of a Üid its rec9mcal is , thn numbers [CDS (I) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_i_math_q26]

(a) 1 only (b) 96" cm2 (c) 2 only (d) oå 1 2 69T cm2

**421.** Consider the following statements : I. The line segment AP divides the area of the triangle ABC into two equal parts [CDS (I) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_i_math_q29]

(a) I and 11 only (b) 11 and Ill only

(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

**422.** If two corresponding sides Of two similar triangles, are in the ratio 9 : 4, then What • the ratio Of their areas ? (c) 81 16 / 26A (d' 27 : [CDS (I) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_i_math_q32]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

**423.** Let ABC be a triangle with AB = 3 Cm, AC = 5 cm. If AD is a median drawn from the vertex A to the side BC, then which one of the following is correct ? [CDS (I) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_i_math_q54]

(a) is always greater than 4 cm but less than 5 cm (b) is always greater than 5 cm

(c) is always less than 4 cm (d) None of the above Consider the following statements

**424.** A's salary was increased by and then decreased by 20%. On the whole A's salary is increased by [CDS (I) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_i_math_q96]

(a) 100066-1 (b) 10000m (c) 100+-6 (d) 24.16 m

**425.** In an election 10% of the voters on the voter list did not cast their vote and 60 voters cast their ballot papers blank. nere were Only two candidates. The winner was supported by 47% of total voters in the voter list and he got 308 votes more than his rival. The number of voters on the voter list is [CDS (I) 2023 : Elementary Mathematics Real Test Paper Qcds_2023_i_math_q97]

(a) 3600 (b) 6200 (c) 6028 (d) 6400

*— SPACE FOR ROUGH WORK —*

**426.** ADM DCM ABCD t, Afa AB-CDtl [CDS (II) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_ii_math_q18]

(a) os x (b) sin X + cos x

(c) sin x — cos x 63. • l. sin r > sin 1 2. cos l' cos I (d) 4q 66. 67.

**427.** AD i s the diameter Of a rcle with area 707m2 and BC- as shown in the figure above. All curves inside the circle are semicirc s with their diameters on AD. What is the cost Of levelling the shaded regio at the rate of 63 per square metre? (d) e 29,700 e 22,400 14,847 None Of the above [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q2]

(a) M < Median (b) GM Median

(c) M = Me dian Suppose for Which one of the corre ct? (d) GM>1/6

**428.** Two circles of radii 20 cm and 16 cm intersect and the length of common chord is 24 cm. If d is the distance between their centres, then which one Of the following is correct ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q3]

(a) < 26 cm (b) 26 cm < d < 27 cm (c) 27 (d) > 28 cm

**429.** The diagonal BD is parallel to the tangent of the circle at A Select the answer using the code given below : [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q6]

(a) I only (b) 11 only (c) oth I (d) Neither I nor

**430.** A parallelogram and a rectangle stand on the same base and on the same side of the base with the same height. If 11, be the perimeters of the parallelogram and the rectangle respectively, then which one of the following is correct ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q7]

(a) 11 (b) 11 = (c) 12 but 11 *212 (d) 212

**431.** The area of a rectangle whose length is 5 more than twice its width is 75 square units. What is the perimeter of the rectangle ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q10]

(a) 40 units (b) 30 units

(c) 24 units (d) 20 units Two number* in the ratio

**432.** The perimeter of a rectangle having area equal to 144 cm and sides in the ratio 4 : 9 is 52 cm (b) 56 (c) 60 cm (d) 64 cm [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q11]

(a) 56 (b) 60 cm (c) 64 cm (d) 1 only

**433.** What is the area between a square Of side IO cm and two inverted semicircular cross-sections each of radius 5 cm inscribed in the square? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q12]

(a) 56 (b) 60 cm (c) 64 cm (d) 1 only

**434.** From a circular piece of cardboard of radius 3 cm, two sectors of4(r each have been cut Off. The area Of the remaining portion is [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q13]

(a) I I square cm (b) 22 square cm

(c) 33 square cm (d) 44 square cm 17. (c) (d) 15. (1

**435.** Three rectangles RI, Ra and Rs have the same area. Their lengths x, , xz 'and x, respectively are such that x, < x, < xy If VI , V2 and V3 are the volumes of the cylinders formed from the rectangles RI, R2 and Rs respectively by joining the parallel sides along the breadth, then which one of the following is correct ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q15]

(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

**436.** What is the cost Of painting per square centimetre ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q17]

(a) Rs. (b) Rs. (c) Rs. (d) Rs.

**437.** In a pi-diagram there are three sectors. If the ratio Of the angles Of the sectors is 1 : 2 : 3, then what is the angle of the largest sector ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q18]

(a) 2000 (b) 1800 (c) 1500 (d) 1200

**438.** A square and a rectangle have equal areas. If one side of the rectangle is of length numerically equal to the square of the length of the side of the square then the other side of the rectangle is [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q19]

(a) square root of the side of the (b) half the side of the square

(c) of unit length (d) ouble the side of the square

**439.** Suppose a region is formed by removing a sector of 200 from a circular region of radius 30 feet. What is the area of this new region ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q20]

(a) 1507t square feet (b) 550z square feet (c) square feet (d) 850% square feet A -

**440.** A persm wishes to fence 375 m2 rectangular garden. He has 65 m of barbed wire and is able to fence only three sides of the garden. What is the perimeter of the garden ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q22]

(a) 80 m (b) 84 m (c) 90 m (d) 100m

**441.** What is the maximum area of a rectangle, in square cm, whose perimeter is 400 cm ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q23]

(a) 100 (b) 200 (c) 1000 (d) 10,000

**442.** In a rectangle ABCD, AC is one of the diagonals. If AC+AB=3AD and AC— AD 4 units, then what is the area of the triangle ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q24]

(a) 24 square unit (b) 36 square unit (c) 48 square unit (d) 72 square unit

**443.** A metallic sheet of rectangular shape is 48 cm x 36 cm. From each corner a square of side x cm is cut off and an open box is made of the remaining sheet. If the box holds 5•12 litre of water, then what is the value of x ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q26]

(a) 6 cm (b) 5 cm (c) 8 cm (d) 5 cm

*— SPACE FOR ROUGH WORK —*

**444.** If three cubic biscuits having edges m, m and 0-5 m respectively are melted and formed into a single cubic biscuit, then What is the total surface area Of the cubic biscuit? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q28]

(a) rn2 (b) 56 rn2 (c) 84 m2 (d) 2•16 m

**445.** A cardboard sheet in the form of a circular sector of radius 30 cm and central angle 1440 is folded to make a cone. What is the radius of the cone ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q29]

(a) 12 cm (b) 18 cm (c) 21 cm (d) None Of the above

**446.** If three metallic spheres Of radii 6 8 cm and IO cm are melted to form a single sphere, then the diameter Of the new sphere will be [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q30]

(a) 12 Cm (b) 24 cm (c) 30 cm (d) 36 Crn

**447.** A large water tank has the shape of a cube. If 128 mJ of water is pumped out, the water level goes down by 2 m. "men the maximum capacity of the tank is [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q31]

(a) 512 m3 (b) 480 ms (c) 324 (d) 256 my

**448.** The height of a cone is 60 cm. A small cone is cut Off at the top by a plane parallel to the base and its volume is -L the volume of original cone. What is the height from the base at which the section is made ? (b) 20 cm (c) 30 cm (d) 45 cm [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q32]

(a) Rs. (b) Rs. (c) Rs. (d) Rs.

**449.** The length, breadth and height Of a cuboid are consecutive integers. If the volume of the cuboid is 336 cubic units, then what is the total surface area Of the cuboid? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q33]

(a) 288 square units (b) 292 square units

(c) 296 square units (d) nnot be determined due to insuffcient data

**450.** The election result in Which six parties contested was depicted by a pie chart. Party A had an angle 135E on this pie chart. If it secured 21960 votes, how many valid votes in total were cast? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q34]

(a) 51240 (b) 58560 (c) 78320 (d) 87840

**451.** If x is the mean Of the numbers in the first pair and y is the mean of the numbers in the second pair, then what is the mean of x and y ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q35]

(a) 60 (b) 72 (c) 75 (d) 78

**452.** The sum of first 47 terms of the series 11 4564 56456 (b) 6 6 9 20 - A 12 34.. Which one of the fellowing is correct ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q37]

(a) (x + 2) is a factor of — + — 24.1 + (b) (x +2) is a factor Of -12r2+24x-32

(c) (x — 2) is a factor Of x' + (d) (x — 2) is a factor of x' + 0-1* +24x-32

**453.** A number consists of two digits, whose sum is [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q38]

(a) 100066-1 (b) 10000m (c) 100+-6 (d) 24.16 m

**454.** Ihe product of digits of the number is (c) 10 (d) [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q39]

(a) 100066-1 (b) 10000m (c) 100+-6 (d) 24.16 m

**455.** From a rectangular sheet of sides 18 cm and 14 cm, a semicircular portion with smaller side as diameter is taken out. Then the area of the remaining sheet will be [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q41]

(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

**456.** What is the value of tanl Otan20tan30tan40 . . tan890 ? [CDS (I) 2022 : Elementary Mathematics Real Test Paper Qcds_2022_i_math_q47]

(a) 150 (b) 300 (c) 450 (d) 600

**457.** What is the number of rounds that a wheel of diameter -N m will make in traversing 7 km ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q2]

(a) 3300 (b) 3500 (c) 4400 (d) 4900

**458.** If sin6+cos9= 2 sing— coso equal to ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q3]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

**459.** M'llat is the value Of x ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q5]

(a) 10 (b) e (c) e (d) 60 (a) (c) (d) 6 9

**460.** The average height of students in the class is 4-6 feet. Ifthe average height of the boys in the class is feet, then what is the average height of the girls in the class ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q6]

(a) Less than 4-2 feet (b) More than 4•2 feet but less than 4-3 feet

(c) More than 4•3 feet but less than 4-4 feet (d) More than 44 feet but less than feet 11 A -

**461.** Let None of the above XYZ be a 3-digit number. Let S-XYZ+YZX+ZW. Which of the following statements Ware correct ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q7]

(a) , B and C (b) , B and D

(c) , E and F 41. The number of three digit numbers (d) Six

**462.** In the given figure, if 6 and = 5, then what is the value of x ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q12]

(a) 450 (b) 300 (c) 150 (d) 100

**463.** pencils, 6 notebooks and 7 erasers cost *250; whereas 6 pencils, 4 note- books and 2 erasers cost [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q16]

(a) only (b) k>2v5 only

(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I—aßy

*— SPACE FOR ROUGH WORK —*

**464.** If O, where then what is 'the value Of 12cot2 B + 3cosece? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q19]
(a) (0+112 (b) (0+2)2 (c) p" +1)2 (d) (30+1)2

**465.** What is (1 + tana tanß)2 + (tana — tanß)2 equal to ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q20]
(a) tanza tan2ß (b) sec2a sec2ß (c) tan2a cot2ß (d) sec2a tan2ß

**466.** What is equal to ? 2 cos30 — cose [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q24]
(a) sun29 (b) os2e (c) ot9 (d) tano A -

**467.** If LBAC— then what is sine + cos9 equal to ? 71 65 73 65 79 (d) [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q25]
(a) I and 11 only (b) 11 and Ill only
(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

**468.** Moidam-The Mound-Burial system of the Ahom dynasty was inscribed in the UNESCO World Heritage lisu The Government Of India conferred the status of classical language to Pali, Assamese, Bengali and Maithili. Project PARI is an initiative Of Ministry of Education Select the answer using the code given below : [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q26]
(a) 2 only (b) I only (c) 2 and 3 only (d) 1,2and3

**469.** What is (p sine + qcos9)equal to? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q30]
(a) mn — I O For the following two (b) 2/3 For the following tvo
(c) 1 P.T.O. For the following two (d) 2/3 sina 4 u sin # cosa cosß

**470.** If 3sine + 4cos9 —5, then what is a value of 4tan0 + 3cot0 ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q32]
(a) 25 (b) 50 (c) 75 (d) 25

**471.** ZPAD+ ZPCB Select the answer using the code given below : [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q34]
(a) I only (b) 11 only (c) oth landll (d) Neither I nor 11 A -

**472.** What is a value of (cosecO + secO) ? 35 12 25 12 For the next two (02) items that follow : cosece — sino = p' and seco — cose = q3 [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q35]
(a) 1/2 (b) 1/√2 (c) √3/2 (d) 1

**473.** What is cot2ß equal to ? p For the next two (02) items that follow : p + q coto = 3cosec0 and q — p cote 2cosec9 [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q36]
(a) 96. What is cot2ß equal to ? p For the next two (b) 10
(c) 13 For the next two (d) 2pq — —

**474.** If p and q are the roots of the equation x2 — sin29x — cos2B 0, then what is the minimum value of p2 + q2 ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q38]
(a) nM — k (b) n (c) 81 (d) 105

**475.** If sin x cos x = 1/2, then what is the value of sin x — cos x ? nt lat of Iftan2 y cosec2 x — 1 tana y, then which 81. one of the following is correct ? (b) x = 2y (c) y — 2x [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q40]
(a) The angle bisector of angle A (b) The angle trisector Of angle A
(c) isector of angle A (d) None of the above What is log

**476.** 22 COS x cos x 2 , which one I + cosec x cosecx — of the following is one Of the values [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q41]
(a) The angle bisector of angle A (b) The angle trisector Of angle A
(c) isector of angle A (d) None of the above What is log

**477.** If sinO+cosO=i, then what is the value of sine dose? 2 (d) 1/2 I + sin a [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q46]
(a) 1/2 47. (b) 60 (c) 45 (d) 30

**478.** What is equal to? I—sin0 (b) secG+ tane (c) cosece + cot é (d) cosec — c Ot [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q47]
(a) 1/2 47. (b) 60 (c) 45 (d) 30

**479.** sin cos (9-20) 6 0 — cos e — cos I — sin2 B cos2 e 2-5 kmph 3aqå 3-25 kmph t I [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q49]
(a) 16 cm (b) 20 em
(c) 25 cm (d) 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ?

**480.** If tan O 3/4 and O is acute, then what is sin 0 equal to ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q55]
(a) I If f (b) -20 (c) 20 (d) 12

**481.** A— sin A What is gin3 A €9) — cos A see A sec A cos A None Of the ain A and is an angle. A equal if Sih following correct ? cos B COS * cos COS 2 sin O cos O 2 sin O' sec B tan b one Of A — sin A [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q56]
(a) 1 (b) √3 (c) 1/√3 (d) 2

**482.** What is equal to? cotx tan x [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q57]
(a) sin x (b) os x (c) tan x (d) ot x

**483.** m 300m 300 Jfä 500 600 4Tä 900 (d) sin x —cos x + I sin X 4-cosx — I sin X — I [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q59]
(a) sin x —cos x + I sin X 4-cosx — I cos x —cos y sin X — I (b) OS X sin x + I cos x y COS X sin x —I sin x —cos y
(c) OS X + I sin x + I (d) sin 2 x — sin y cosx+ i IA 59. 60.

*— SPACE FOR ROUGH WORK —*

**484.** DCM + ABM ABCD e I 61. (l t? (e) sin x (d) cos x (cosecx — sin x) (sec x — cos x) (tan x + cot* ) [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q61]

(a) os x (b) sin X + cos x

(c) sin x — cos x 63. • l. sin r > sin 1 2. cos l' cos I (d) 4q 66. 67.

**485.** The value of -loglo g10 2 32 5 is If tane + cote , where O < < then 2 sin + cos 9 is equal to 0-1 (c) 2 / 54A [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q64]

(a) M < Median (b) GM Median

(c) M = Me dian Suppose for Which one of the corre ct? (d) GM>1/6

**486.** If sin x + sin2x=l, then what is the value of cossx + 2cos6x + cos x ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q65]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

**487.** If tan2x+ = 2 and 00 < x < 900, tan x then what is the value of x ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q66]

(a) 150 (b) 300 (c) 450 (d) 600

**488.** Consider the following : COS 750 sin 150 cos 350 2. sin 550 sin 800 3. cos sin 120 cos 180 cos 780 sin 720 sin 1 10 cos 790 cos 280 cosec 620 = 1 —sin 590 sec 310=0 Which Of the above are correct ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q67]

(a) 1 and 2 only (b) 2 and 3 only (c) I and 3 only (d) 1, 2 and 3 - A

**489.** cosa + cosß 2 3m sina sinß- O t, Osa s 9C, O S ß S 909 cosm -cos2ß t? 56. sece+cosB-—, os0S900 t, sin 2 e t? 2 3 57. (1+cot8 t sece) 4 [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q69]

(a) 11 (b) 22

(c) 77 (d) 99 k sece 59. 60' 60. 60• 45'

**490.** What is cos9—2cos3e' equal to ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q71]

(a) sin (b) ose (c) tan (d) ot O

**491.** What is the value of the following? 2 sin 680 2cot 150 cos 220 5 tan 750 3 tan20' tan4cr tan 450 tan 5 The perpendicular dropped from a vertex 12. of a right-angled triangle the hypotenuse divides it into two segments Of lengths 9 units and 16 units respectively. What is the length Of the perpe n dicular? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q72]

(a) 6 units (b) 8 units (c) IO units (d) 12 units

**492.** tan (3A) (A - 220), SA W [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q74]

(a) 250 (b) 270

(c) 280 (d) 300 sin e — Cos + I z p sec 9 + q tan O, sin cos 54. :

**493.** Two towers A and B of height 23 m and 11 m respectively, stand 9 m apart. A straight rod is joined to the two tops of the towers. A monkey sitting on the top of A, climbs the rod to reach the top of B. If the monkey takes 5 minutes to reach the other end, what is the average speed of the monkey ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q75]

(a) IO mimin (b) 5 mimin (c) 10 cm/sec (d) 5 cm/sec

**494.** What is (sece-tan9) to? (b) mane (d) sine + cos B I — sin I + sin 9 6, then What is as. If cote coec2e — sec2 e coxc29+sec e equal to? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q76]

(a) 1/2 (b) 1/3

(c) 2/3 (d) 3/4 difference between the two acute

**495.** What is p2q2(p2 + q2 + 3) equal to? For the following two (02) items : (d) 2/3 sina 4 u sin # cosa cosß [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q77]

(a) mn — I O For the following two (b) 2/3 For the following tvo

(c) 1 P.T.O. For the following two (d) 2/3 sina 4 u sin # cosa cosß

**496.** What is tan 2 equal to? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q78]

(a) mn — I O For the following two (b) 2/3 For the following tvo

(c) 1 P.T.O. For the following two (d) 2/3 sina 4 u sin # cosa cosß

**497.** What is tan 2 ß equal to? 1/3 For the following tvo (02) items : I + sin e COS [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q79]

(a) mn — I O For the following two (b) 2/3 For the following tvo

(c) 1 P.T.O. For the following two (d) 2/3 sina 4 u sin # cosa cosß

**498.** At a point on level ground, the tangent of the angle of elevation of the top of a tower is found to be S. On walking 70 m towards the tower, the tangent of the angle of elevation of the top of the tower is found to be Y. What is the height of the tower ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q81]

(a) 25 (b) 50 (c) 75 (d) 25

**499.** What is tane equal to ? (29+3p) [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q82]

(a) 96. What is cot2ß equal to ? p For the next two (b) 10

(c) 13 For the next two (d) 2pq — —

**500.** If A, B and C are interior angles of a triangle ABC, then what is equal to ? 2 A+B+C (c) sin A+B+C (d) tan 4 In a triangle ABC, right-angled at B, AB + BC + cm and length of the hypotenuse is 20 cm. What is the value of tanA + tan C ? 4 2 (d) 20 17 the [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q88]

(a) sin (b) ose (c) tan (d) ot O

**501.** If p, Q and R are the areas of sectors at A, B and C within the triangle respec- tively, then which of the following is/are correct ? I. P = Æcm2 [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q89]

(a) 10•05 cm2 (b) 10•15cm2

(c) 05 cm2 (d) 11•15cm2 Consider the following for the next three

*— SPACE FOR ROUGH WORK —*

**502.** If AD-5cm, BD-4cm and 6 cm, then what is AB equal to ? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q90]
(a) 10•05 cm2 (b) 10•15cm2
(c) 05 cm2 (d) 11•15cm2 Consider the following for the next three

**503.** In a triangle ABG AD is the bisector of ZBAC If 12 em, IO cmand DC — 5 cm. then what is the perimeter of the triangle? [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q91]
(a) 30 cm (b) 31 cm (c) 33 cm (d) 35 cm

**504.** The angles in a right-angled triangle is — radian. One of the acute angles Of the triangle is [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q93]
(a) 600 (b) 50 (c) 50 (d) 50

**505.** If ABC is a triangle right angled at C and having u units, v units, w units as the lengths of its sides opposite to the veltices A, B, C respectively, then what is the tangent of the angle at A + tangent of the angle at B equal to ? (d) [CDS (II) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_ii_math_q98]
(a) 4 days (b) 5 days (c) 6 days (d) ys

**506.** How many such sequences of five consecutive integers are possible ? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q38]
(a) One (b) Two (c) Three (d) Five

**507.** Question : What is the cost of 15 pens, 21 pencils and 18 note books ? The cost of 7 pens, 6 pencils and 5 note books is [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q39]
(a) 15° (b) 30° (c) 45° (d) 60°

**508.** What is tang equal to ? (C) pq [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q43]
(a) 15° (b) 30° (c) 45° (d) 60°

**509.** What is tan2a equal to ? (b) [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q44]
(a) 96. What is cot2ß equal to ? p For the next two (b) 10
(c) 13 For the next two (d) 2pq — —

**510.** D 3k E BC 3 ADE 3 cm2 I EC ? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q48]
(a) 16 cm (b) 20 em
(c) 25 cm (d) 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ?

**511.** What is 5% of of 500 ? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q51]
(a) 12.5 (b) 25 (c) 1.25 (d) 6.25

**512.** Ai if one Of the Of the — bx + c O the square at the then Which one the following is ? 3bc c c Sbe + b [d) 3be 4 b3 + then what is x What ig the value Of O ne Ot factors tf — 5x2 — 81 ia ta) equgi to ? Let A : % is odd integer) and B — : xa — 8x + % — [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q52]
(a) 5% (b) 8% (c) 10% (d) 12.5%

**513.** 1%en which one of the ig ? (b) (41 ACB sum Of numbers 7 and sum Of 20. their ' is [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q53]
(a) Rs. 1,200 (b) Rs. 1,440 (c) Rs. 1,500 (d) Rs. 1,800

**514.** A person selling an article for 96 finds that his loss percent is one-fourth Of the amount Of rupees that he paid for the article. What can be the cost price? (d) e 160 only e 240 only e 160 or 240 Neither e 160 nor 240 If (x + k) is the common factor of x + b and x2 + d, then What is k equal to? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q54]
(a) 125 (b) -3125
(c) 3125 IA 46. (d) e 160 only e 240 only e 160 or 240 Neither e 160 nor 240

**515.** He sold the chair at a gain of 5% and the table at a gain of 20%. He gained 16% on the whole. What is original cost of table ? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q58]
(a) *400 (b) *450 (c) V550 (d) e600

**516.** The ratio of the surface areas of the spherical and cylindrical portions is - A [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q61]
(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

**517.** What is the ratio of the interest on the first loan to the interest on the second loan ? For the next two (02) items that follow : Let 12(tane + cot0) = 25, where 450 < 0 < [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q62]
(a) 1/2 (b) 1/√2 (c) √3/2 (d) 1

**518.** A car is traveling at a constant rate of 45 km per hour. The distance travelled by car from 10.40 am to 1.00 pm is [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q63]
(a) 165 km (b) 150 km
(c) 120 km (d) 105 km A number consists of two digits whose sum is

**519.** A and B are two taps which can fill a tank individually in 10 minutes and 20 minutes respectively. However, there is .a leakage at the bottom, which can empty a filled tank in 40 minutes. If the tank is empty initially, how much time •c will both the taps take to fill the tank with leakage ? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q64]
(a) 2 minutes (b) mihutes (c) 5 minutes (d) 8 minutes

**520.** An aeroplane flying at a height of 3000 m passes vertically above another aeroplane at an instant when the angles of elevation of the two planes from some point on the ground are 600 and 450 respectively. Then the vertical distance between the two planes is [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q67]
(a) 100066-1 (b) 10000m (c) 100+-6 (d) 24.16 m

*— SPACE FOR ROUGH WORK —*

**521.** A tank can be filled by pipe X in 2 hours and pipe Y in 6 hours. At IO a.m. pipe X was opened. At what time will the tank be filled if pipe Y is opened at 11 a.m. ? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q71]

(a) 12:45 hours (b) 5:00 p.m. (c) 11:45 a.m. (d) 11:50 a.m.

**522.** In covering certain distance, the average speeds of X and Y are in the ratio 4 : [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q72]

(a) , B and C (b) , B and D

(c) , E and F 41. The number of three digit numbers (d) Six

**523.** Two circles touch externally. The sum of their areas is 411t square cm. If the distance between their centres is 9 cm, then what is difference between their diameters ? (b) I •5 cm (c) 2 cm [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q75]

(a) sin (b) ose (c) tan (d) ot O

**524.** From a rectangular metal sheet Of sides 25 cm and 20 em, a circular sheet as large as possible is cut-off. What is the area of the remaining sheet ? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q79]

(a) 186 (b) 144 (c) 93 cm2 (d) 72 cm2

**525.** If a quantity y varies as the sum of three quantities of which the first varies as x, the second varies as —x+x2, the third varies as x3 —p then what is y equal to ? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q80]

(a) where k is a constant (b) kx+lx2+nu3, where k, l, m are constants

(c) kP where k is a constant (d) kx where k is a constant - A

**526.** What is the „ + sin275ø ? + sin2250 + [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q81]

(a) tan2150 + + tan2250 tan2750 (b) os2150 + cos220e + cos2250 . cos2750

(c) + cot225• — + cot2750 (d) sec2200 + sec2250

**527.** What is the sum of the squarés of all the six numbers ? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q84]

(a) 40 (b) 42 (c) 45 (d) 48

**528.** What are the factors Of [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q85]

(a) (x —2y—4) and (x —2Y+2) (b) (x —y and (x —4 y + 4) (c) (x —y and (x —4 y —4) (d) (x (x +2Y+2) 26k /

**529.** Which one Of the following is a non-termmating and repeating decimal ? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q86]

(a) 13/8 (b) 3/16 (c) 3/11 (d) 137/25

**530.** ABC BC D ZACD- 1200, ZABC- -2? [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q87]

(a) 47. 48. 49. 14 sm I — sine (b) 9 — tan (c) sece+ tan (d) osece + cot

**531.** A number consists of two digits. The sum of the digits is IO. On reversing the digits of the number, the number decreases by [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q89]

(a) 21 (b) 24

(c) 36 (d) 42 Which one among the following is not correct ?

**532.** If tano + = 2, then tang is equal to 3 4 5 4 2 5 (d) 2 - A [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q95]

(a) 100066-1 (b) 10000m (c) 100+-6 (d) 24.16 m

**533.** Ifa < b, then — Select th e correct answer sing the code given below. (b) 2 only (c) 3 only (d) 2 and [CDS (I) 2021 : Elementary Mathematics Real Test Paper Qcds_2021_i_math_q100]

(a) M < Median (b) GM Median

(c) M = Me dian Suppose for Which one of the corre ct? (d) GM>1/6

**534.** 413283P759387, 13 t, 2. 21000000 7 OTT? /50A 4. 5. 6. 50 100 ('L n) 2 5 (al +a2 + + b2 + b3 + b4) [CDS (II) 2020 : Elementary Mathematics Real Test Paper Qcds_2020_ii_math_q1]

(a) 15 (b) 30 (c) 45 (d) 60 2727 -1527*6@avråa OTT?

**535.** The shadow Of a tower is 15 m when the Sun's altituqe is [CDS (II) 2020 : Elementary Mathematics Real Test Paper Qcds_2020_ii_math_q31]

(a) 1/2 47. (b) 60 (c) 45 (d) 30

**536.** What is the least value of (25 cosec2x + sec2x) ? [CDS (II) 2020 : Elementary Mathematics Real Test Paper Qcds_2020_ii_math_q54]

(a) 40 (b) 36

(c) 26 (d) 24 and If a cot n9, then which one of following is correct ?

**537.** If cos I —sina cos O = 4, then which I + Sin one of the following is a value of (tan29 + cot29) ? 10 17 (b) (d) [CDS (II) 2020 : Elementary Mathematics Real Test Paper Qcds_2020_ii_math_q65]

(a) 25 (b) 50 (c) 75 (d) 25

**538.** Chord CD intersects the diameter AB of a circle at riW1t angle at a m)int p in the ratio I : [CDS (II) 2020 : Elementary Mathematics Real Test Paper Qcds_2020_ii_math_q86]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

**539.** A pole is standing erect on the ground which is horizontal tip of the_pole is tied tight with a roly of length VI 2 m to a point on the ground. If the rope is making 300 with the horizontal, then the height of the pole is [CDS (I) 2020 : Elementary Mathematics Real Test Paper Qcds_2020_i_math_q76]

(a) 100066-1 (b) 10000m (c) 100+-6 (d) 24.16 m

**540.** 12288, 28200 44333 (qqFT) 15. = (b) 1000 (c) 100000 (d) 1000000 [CDS (II) 2019 : Elementary Mathematics Real Test Paper Qcds_2019_ii_math_q16]

(a) 272 (b) 1000 (c) 100000 (d) 1000000

*— SPACE FOR ROUGH WORK —*

**541.** A tank can be filled by pipe Xin 2 hours and pipe Y in 6 hours. At 10 a.m. pipe X was opened. At what time will the tank be filled if pipe Y is opened at 11 a.m. ? [CDS (I) 2019 : Elementary Mathematics Real Test Paper Qcds_2019_i_math_q10]

(a) 12:45 hours (b) 5:00 p.m.

(c) 11:45 a.m. (d) 11:50 a.m. qÄ a b aG+bdj=32

**542.** If its parallel sides are in the ratio 5 : 7 and the perpendicular distance between them is 14 cm, then the smaller of the parallel sides is [CDS (I) 2019 : Elementary Mathematics Real Test Paper Qcds_2019_i_math_q26]

(a) 20 cm (b) 22 cm (c) 24 cm (d) 26 cm

**543.** The age of a woman is a two-digit integer. On reversing this integer, the new integer is the age of her husband who is elder to her. The difference between their ages is one-eleventh of their sum. What is the difference between their ages ? [CDS (I) 2019 : Elementary Mathematics Real Test Paper Qcds_2019_i_math_q99]

(a) 8 years (b) 9 years (c) 10 years (d) 11 years

**544.** The number of values of x satisfying 100 x + — > 50, where x is a natural number less than or equal to 100 is [CDS (II) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_ii_math_q50]

(a) 51 (b) 53 (c) 55 (d) 57

**545.** Two trains, one is of 121 m in length at the speed of 40 km/hour and the other is of 99 m in length at the Speed of 32 km/hour are running in opposite directions. In how much time will they be completely clear from each other from the moment they meet ? [CDS (II) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_ii_math_q64]

(a) IOS (b) Ils (c) 16 s (d) 21 s

**546.** Which one of the following is correct ? [CDS (I) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_i_math_q7]

(a) ecimal expansion of a rational number (b) ecimal expansion of a rational number

(c) ecimal expansion (d) ecimal expansion of an

**547.** If a ^ - l = 5, then what is the value of a^ [CDS (I) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_i_math_q9]

(a) 125 (b) -125 (c) 140 (d) -140

**548.** The smallest integer with 4 digits which is a perfect square is [CDS (I) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_i_math_q12]

(a) 1000 (b) 1024 (c) 1089 (d) None of the above

**549.** What is the value of k ? [CDS (I) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_i_math_q16]

(a) - (b) 14 (c) - (d) 7

**550.** a - c , - c l + x^-'^ +X where x 0 ? [CDS (I) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_i_math_q20]

(a) - (b) 0 (c) 1 (d) 3

**551.** appear for the examination, — of those who 20 appeared passed. If the number of registered students is 2000, the number who passed is [CDS (I) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_i_math_q23]

(a) 1920 (b) 1056 (c) 1020 (d) 864

**552.** If A : B = 1 : 2, B : C = 3 : 4, C : D = 2 : 3 and D : E = 3 : 4, then what is B : E equal to ? [CDS (I) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_i_math_q24]

(a) 3 :2 (b) 1: (c) 3 :8 (d) 4 :1

**553.** ^ 120 is distributed among A, B and C so that A's share is ^ 20 more than B's and W 20 less than C's. What is B's share ? [CDS (I) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_i_math_q30]

(a) 10 (b) 15 (c) 20 (d) 25

**554.** What is the original amount that the merchant invested ? [CDS (I) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_i_math_q33]

(a) ?= 5,120 (b) ^ 5,210 (c) ^ 5,350 (d) ^ 5,500

**555.** For 0 < m < 1, which one of the following is correct ? [CDS (I) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_i_math_q36]

(a) ■ logj^Q m < m^ < m < m“ ^ (b) m < m~^ < m^ < logjQ m

(c) logio ni< m < m“ ^ < m^ (d) • logj^Q m < m“ ^ < m < m^

**556.** A gentleman left a sum of 39,000 to be distributed after his death among his widow, five sons and four daughters. If each son receives 3 times as much as a daughter receives, and each daughter receives twice as much as their mother receives, then what is the widow's share ? [CDS (I) 2018 : Elementary Mathematics Real Test Paper Qcds_2018_i_math_q37]

(a) ^ 1,000 (b) ^ 1,200 (c) 1,500 (d) None of the above

**557.** If p and q (p > q) are the roots of the equation xa — 60x + 899 = O, then which one of the following is correct ? -24+27=0 (c) 2p—q 30 0 (d) 3p—2q 43 0 A - [CDS (II) 2016 : Elementary Mathematics Real Test Paper Qcds_2016_ii_math_q2]

(a) only (b) k>2v5 only

(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I—aßy

**558.** If the roots of the equation x2—4x— O are real, then what is the minimum value of N? [CDS (II) 2016 : Elementary Mathematics Real Test Paper Qcds_2016_ii_math_q3]

(a) 1 (b) 01 (c) 001 (d) 0001

**559.** If a and are the roots Of the equation then what is equal to? [CDS (II) 2016 : Elementary Mathematics Real Test Paper Qcds_2016_ii_math_q6]

(a) 16 (b) 25 (c) 36 (d) 49 1 P.T.O.

**560.** What was the original wage bill of the factory if the present bill is Rs. 5,000/- ? (b) Rs. 6000/- (c) Rs. 8,000,'- (d) None of the above [CDS (II) 2016 : Elementary Mathematics Real Test Paper Qcds_2016_ii_math_q16]

(a) 4 days (b) 5 days (c) 6 days (d) ys

**561.** If a and are the two zeros of the polynomial 25x2 — 15x + 2, then what is a quadratic polynomial whose zeros are (2a)-l and (2ß)-l ? [CDS (II) 2016 : Elementary Mathematics Real Test Paper Qcds_2016_ii_math_q17]

(a) x2+30x+2 (b) 8x2-30X+25 (c) 8x2-30x (d) x2+30x

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**562.** Rmi Five yen back, the of Radha Was tima that of Rni, but buk age of WE pf Rmi. is age them' ? (c) 10 (d) [CDS (II) 2016 : Elementary Mathematics Real Test Paper Qcds_2016_ii_math_q18]

(a) 1 only (b) 96" cm2 (c) 2 only (d) oå 1 2 69T cm2

**563.** Which one of the following is the largest number among 22222, 22222, 22222, 22222 ? [CDS (II) 2016 : Elementary Mathematics Real Test Paper Qcds_2016_ii_math_q21]

(a) prime number (b) not a prime, but power of a prime

(c) not a power of a prime, but a composite even number (d) not a of a prime, but a composite odd number

**564.** Which one of the following equations does not have real roots ? (b) 2r2+10x-1-o (d) •4x2 + A - [CDS (II) 2016 : Elementary Mathematics Real Test Paper Qcds_2016_ii_math_q28]

(a) 29, 17, (b) 31, 17, (c) 19, 37, (d) 23, 31,

**565.** What is the value of sin2150 + sin2200 + sin2250 + . . + sin2750 ? [CDS (II) 2016 : Elementary Mathematics Real Test Paper Qcds_2016_ii_math_q94]

(a) tan2150 + tan2200 + tan2250 + tan2750 (b) os2159 + cos2200 + cos2250 cos2750

(c) ot2150 — cot2200 + cot2250 — + cot2750 (d) sec2150 + sec2200 + sec2250 + sec2750

**566.** If x then the value of [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q8]

(a) 35 (b) 40 (c) 45 (d) 50

**567.** X can do a piece of work in 4 days and Y can do the same piece of work in 6 days. Which of the following statements is/ are correct ? I. If X and Y work alternately starting with X, then the piece of work will be finished on the fifth day. [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q23]

(a) 3 months (b) 6 months (c) 9 months (d) 1 year

**568.** If the roots of the quadratic x2 —4x— log10N=O are all real, minimum value of N is [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q25]

(a) .100 (b) 1000 (c) 10000 (d) 10000 equation then the

**569.** In the figure given above, AB is parallel to CD. If Z DCE = x and Z ABE = y, then what is Z CEB equal to ? (c) x + y — (11/2) 980 1440 Iri the figure given above, AB is parallel to CD. If Z BAF 980 and AFC = 1440, then what is L ECD equal to ? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q28]

(a) 620 (b) 640 (c) 820 (d) 840

**570.** Which of the points P(5, — I), QI 3, —2) and R(I, I) lie in the solution of the system Of inequations x + y 4 and x-y>2? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q30]

(a) Q and R o nly (b) P and R o nly (c) P and Q o Illy (d) P, Q and R

**571.** If Of fre lalgths of is 12 what is arn of ? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q34]

(a) 9 an-2 (b) 12 cm2 (c) 16 (d) 25 cma C -

**572.** If one Of a aingle (with all sides intega•s) is 15 an, thm maximum Of uiangle ? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q35]

(a) 240 cm (b) 225 (c) 113 cm (d) 112 cm

**573.** If p is relatively to of g r, p is relatiüely prime to poduct qr. [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q36]

(a) 1 only (b) 96" cm2 (c) 2 only (d) oå 1 2 69T cm2

**574.** What is p2 + q2 equal to ? (d) 13 (c) [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q43]

(a) 96. What is cot2ß equal to ? p For the next two (b) 10

(c) 13 For the next two (d) 2pq — —

**575.** Which one Of the following is correct ? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q44]

(a) 96. What is cot2ß equal to ? p For the next two (b) 10

(c) 13 For the next two (d) 2pq — —

**576.** If the sum of the roots of ax + bx+c—O is equal to the sum of the squares of their reciprocals, then which one Of the following relations is correct? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q46]

(a) 2 = 2a2c (b) 2+ 2b 2 a (c) 2+ bc 2' a 2c (d) 2+b2+c2=I

**577.** Let p be the area of a square X and q be the area of the square formed on the diagonal of the square X. What is the value of — ? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q50]

(a) sin (b) ose (c) tan (d) ot O

**578.** A square is drawn inside a square of side 14 cm in such a way that the corners of the inner square coincide with the mid points of the sides of the outer square. What is the area lying between the two squares ? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q51]

(a) 98 square cm (b) 56 square cm (c) 49 square cm (d) 24•5 square cm

**579.** Consider the following statements . [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q54]

(a) The triangle is isosceles (b) The triangle is equilateral

(c) None Of the ab ove For the next three (d) 3124

**580.** A runs I — times as fast as B. If A gives B a start of 80 m, how far If must the winning post from the starting point be so that A and B might reach it at the same time? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q55]

(a) 200 m (b) 300 m (c) 270 m (d) 160 m

**581.** Sheela can stitch a suit in 2 days. while Meena can stitch a suit in (hys. How many days will both take in stitching 30 suits ? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q60]

(a) 32 days (b) 33 days

(c) 35 days (d) 40 days At the end of the time he received 11,

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**582.** How many days did he work ? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q61]
(a) 20 (b) 21 (c) 24 (d) 25

**583.** N is by 17, is equal to 182. ne b«wen the quotiai md is 175. is åe value of N ? [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q72]
(a) 2975 (b) 3094 (c) 3101 (d) 3269

**584.** What is the value of q ? next two (02) items that For the follow : 16 + [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q79]
(a) 487 (b) 483 (c) 481 (d) 480

**585.** What is 65 8 61 8 59 For the follow : -257-0 equal to ? equal to ? next two (02) items that and y = Let x [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q80]
(a) 487 (b) 483 (c) 481 (d) 480

**586.** What is 0.242424.... [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q89]
(a) 23/99 (b) 8/33 (c) 7/33 (d) 47/198 equal to ?

**587.** The price of an article is Rs. [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q91]
(a) x- (b) x =
(c) x = 11.5 If 0.764 y = 1.236 x, Y—x value Of (d) 0.764

**588.** What is the value of [CDS (II) 2015 : Elementary Mathematics Real Test Paper Qcds_2015_ii_math_q92]
(a) x- (b) x =
(c) x = 11.5 If 0.764 y = 1.236 x, Y—x value Of (d) 0.764

**589.** Jails with highest and smallest percent- age of trained convicts are respectively [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q7]
(a) F and D (b) nd F (c) nd A (d) nd A

**590.** In a race of 100 m, A beats B by 4 m and A beats C by 2 m. By how many metres (approximately) would C beat B in another 100 m race assuming C and B run with their respective speeds as in the earlier race ? (b) 204 (c) 208 (d) 3-2 [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q11]
(a) 100066-1 (b) 10000m (c) 100+-6 (d) 24.16 m

**591.** Jails from which more than half of the trained convicts are offered jobs, are [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q16]
(a) , B and C (b) , B and D
(c) , E and F 41. The number of three digit numbers (d) Six

**592.** A car travels from A to B at a speed Of 40 km/hr, travels back from B to A at a speed of 30 km/hr and again goes from A to B at a speed of 60 km/hr. What is the average speed of the car? [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q21]
(a) km/hr (b) 42 km/hr (c) 40 km/hr (d) km/hr

**593.** The sum of the deviations of a set of n numbers Xl, x2, x3 , x meas- ured from 15 is —90 and the sum of the deviations of the same numbers measured from —3 is [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q32]
(a) 25 (b) 50 (c) 75 (d) 25

**594.** If a quadrilateral has an Inscribed then the sum of a pair Of opposite sides equals [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q35]
(a) Half the sum of the diagonals (b) Sum of the other pair of opposite sides
(c) Sum Of two adjacent sides (d) None of the above

**595.** What (X2 — Y2) equal to ? (d) 8800 [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q48]
(a) 487 (b) 483 (c) 481 (d) 480

**596.** A has avaage of 510 on md 240 days. What is the mnnb« Of visitors in a of 30 days b%inning Snu•y ? [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q60]
(a) 276 (b) 282 (c) 285 (d) 375 C -

**597.** A locomotive engine can go 40 km/hr. Its speed gets reduced by a quantity that varies directly as the square root of the number of wagons attached. It is known that its speed becomes 16 km/hr if 36 wagons are attached. What is the smallest number of wagons with which the engine is unable to move ? [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q65]
(a) 100 (b) 81 (c) 64 (d) 49

**598.** What is to ? 62 +72 +8 7 + 46 — 2 + 92 + 4 + 26 = z + x and then what is the value of - A [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q67]
(a) x = 100 (b) x = IO (c) 330 (d) 340

**599.** Ifp+q+r=0, then what is equal to? (c) 22 (d) [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q70]
(a) -23 (b) -17 (c) 17 (d) 23 which —5k +

**600.** What is the value of k for which (k2— 5k + 4)x2 + (k2 —3k —4)x+ (k2— 4k) 0 is an identity ? 152 [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q71]
(a) -23 (b) -17 (c) 17 (d) 23 which —5k +

**601.** If then what is the value of a4 + b4 + ca equal to ? [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q72]
(a) 2b2 + + c2a2 (b) 2(a2b2 c2a2) (c) 3(a2b2+ Fc2 + c2a2) (d) 4(a2b2 b2c2 + c2a2)

**602.** If—-L — + , then what is pq x + p x equal to ? Fi¯i ¯ —q x—p x —q [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q73]
(a) -23 (b) -17 (c) 17 (d) 23 which —5k +

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**603.** How many square metres Of canvas are used ? [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q80]
(a) 14450 (b) 14480 (c) 14580 (d) 14850

**604.** A ladder of 17 ft. length reaches a window Which is 15 ft. above the ground on one side of the street. Keeping its foot at the same point the ladder is turned to the other side of the street and now it reaches a window 8 ft. high. What is the Width of the street ? [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q84]
(a) 23 ft (b) 15 ft. (c) 25 ft. (d) 30ft.

**605.** If k is any even positive integer, then (k 2 + 2k) is [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q85]
(a) ivisible by (b) ivisible 8 but may no t be divisible
(c) ivisible by 4 but may not be divisible by (d) ivisible by 2 but may not be divisible by

**606.** I APB Cpl) i A, B, C [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q86]
(a) 121-7 121-75 (b) 121-75 ABC t, X Y AX.AYur (c) 16 (d) 20 cm

**607.** Consider the following statements 64. [CDS (II) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_ii_math_q88]
(a) 36 (b) 44 (c) 52 64. (d) 150

**608.** A passenger train and a goods train are running in the same direction on parallel railway tracks. If the passenger train now takes three times as long to pass the goods train, as when they are running in opposite directions, then what is the ratio of the speed of the passenger train to that of the goods train ? (Assume that the trains nm at uniform speeds) [CDS (I) 2014 : Elementary Mathematics Real Test Paper Qcds_2014_i_math_q87]
(a) 2 :1 (b) 3 ; (c) 4 :3 (d) 1 :

**609.** The quantity of water to be added to the mixture to make the • ratio of milk and water 1 : 2 is [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q14]
(a) 30 (b) 32 (c) 33 (d) 35

**610.** A boy saves Rs. 4•65 daily. What is the least number of days in which he will be able to save an exact number of rupees ? [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q18]
(a) 10 (b) 20 (c) 21 (d) 25

**611.** In the figure given above YZ is parallel to MN, xy is parallel to LM and XZ is parallel to LN. Then MY is • [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q34]
(a) the median of triangle LMN (b) the angular bisector of ZLMN
(c) perpendicular to LN (d) perpendicular bisector Of LN

**612.** Which one of the following is an infinite set ? [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q38]
(a) {x: x is a whole number less than or equal to 1000} (b) {x: x is a natural number less than 1000}
(c) {x: x is a positive integer less than or equal to 1000} (d) {x: x is an integer and less than 1000}

**613.** What is Wc e equal to? (c) 1 P.T.O. [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q47]
(a) mn — I O For the following two (b) 2/3 For the following tvo
(c) 1 P.T.O. For the following two (d) 2/3 sina 4 u sin # cosa cosß

**614.** What is p4q2 + p2q4 equal to ? A - For the next two (02) items that follow : A person borrows a sum Of e 16,000 on simple interest at the beginning of a year. Aner 4 months, *24,000 more is borrowed at a rate of interest double the previous one. At the end of the year, the total interest on both the loans is *4800. [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q55]
(a) 4 : (b) 5 : (c) 3 : (d) 2 :

**615.** There are three brothers. The sums of ages of two Of them at a time are 4 years, 6 years and 8 years. The age difference between the eldest and the youngest is [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q58]
(a) 3 years (b) 4 years (c) 5 years (d) 6 years

**616.** If e —y + z, = z + x and then what is the value of - A [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q60]
(a) x = 100 (b) x = IO (c) 330 (d) 340

**617.** The Of y if one Of the Of the — bx + c O the square at the then Which one the following is ? [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q68]
(a) 120 cm³ (b) 150 cm³ (c) 180 cm³ (d) 216 cm³

**618.** What is the value Of 13th observation ? [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q69]
(a) 20 (b) 23 (c) 32 (d) 40

**619.** Which one of the following is a corect statement ? 13, What is the least integral value of k for which the equation has both the roots positive ? [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q70]
(a) 3 years (b) 4 years (c) 5 years (d) 6 years

**620.** What is the least integral value of k for which the equation has both the roots positive ? [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q71]
(a) 3 years (b) 4 years (c) 5 years (d) 6 years

**621.** The shadow of a pole 6 metre high is 15 metre long and at the same time the shadow of a tree is 25 metre long. What is the height of the tree ? [CDS (II) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_ii_math_q95]
(a) 21 m (b) 10m (c) 35 m (d) None Of the above

**622.** If x : y = 7 : 5, then what is (5x— equal to ? [CDS (I) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_i_math_q22]
(a) 5/4 (b) 6/5 (c) 25/31 (d) 31/42 40. 41. 42.

**623.** priucipgl which a simple interest of 12. 55 will be obtained after 9 months the rate Of 3 — % per is (c) e 2,000 (d) 2,500 The expression gives [CDS (I) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_i_math_q30]
(a) natural number (b) n integer and not a number (c) e 2,000 (d) 2,500 The expression gives

*— SPACE FOR ROUGH WORK —*

**624.** If 20 kg of alloy A, 28 kg of alloy B and some more pure copper are mixed to form a third alloy C which now contains copper and tin in the ratio of 6 ; 7, then what is the quantity of pure copper mixed in the alloy C ? [CDS (I) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_i_math_q89]

(a) 3 kg (b) 4 kg (c) 5 kg (d) 7 kg

**625.** The average of the ages of 15 students in a class is 19 years. When 5 new students are admitted to the class, the average age of the class becomes 18-5 years. What is the average age of the 5 newly admitted students ? [CDS (I) 2013 : Elementary Mathematics Real Test Paper Qcds_2013_i_math_q90]

(a) 17 years (b) 17*5 years (c) 18 years (d) 5 years

**626.** If + px2 + qx + r is an integer for all integral values of x, then consider the following statements : I. p must be an integer 11. q must be an integer Ill. r must be an integer Which of the statements given above is/are correct ? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q24]

(a) I and 11 only (b) Ill only

(c) I, IIandIII (d) None of the statements is correct A -

**627.** If 0.764 y — 1.236 x, then what is the value of [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q25]

(a) 0.764 (b) x = (c) x = (d) x = 11.5

**628.** A man buys a television set which lists for Rs. 5,000 at 10% discount. He gets an additional 2% discount (after the first discount) for paying cash. What does he actually pay for the set ? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q27]

(a) Rs. 4,410 (b) Rs. 4,400 (c) Rs. 4,000 (d) Rs. 4,500

**629.** In which one of the following cate- gories of employees, the percentage change in number of employees in 2020 was maximum as compared to 2019 ? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q30]

(a) 101 •39 cm2 (b) 111-39cm2

(c) 121 •39cm2 (d) 131-39cm2 Consider the following for the next two

**630.** What was the percentage increase of category-E employees in 2020 as compared to that in 2019 ? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q31]

(a) 101 •39 cm2 (b) 111-39cm2

(c) 121 •39cm2 (d) 131-39cm2 Consider the following for the next two

**631.** What are the two categories ? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q32]

(a) E and F (b) nd E (c) nd E (d) nd C A -

**632.** What is the percentage of copper in (b) 64% [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q34]

(a) 487 (b) 483 (c) 481 (d) 480

**633.** Four years ago, the average age of A and B was 18 years. Now the average age Of A, B and C is 24 years. After 8 years, the age of C will be [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q38]

(a) 32 years (b) 28 years (c) 36 years (d) 40 years - A (a) (b) (c)

**634.** Which of the above are correct? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q41]

(a) I and 11 only (b) 11 and Ill only (c) I and Ill only (d) 1, 11 and 111

**635.** What is OQ equal to ? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q52]

(a) 5 cm (b) 6 cm 59. What is PM equal to ?

(c) IO cm (d) 12 cm

**636.** What is PM equal to ? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q53]

(a) 5 cm (b) 6 cm 59. What is PM equal to ?

(c) IO cm (d) 12 cm

**637.** The inner and outer radii of a 7 m long hollow iron right circular cylindrical pipe are 2 cm and 4 cm respectively. If 1000 cni3 of iron weighs 5 kg, what is the weight of the pipe ? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q66]

(a) 264 kg (b) 132 kg (c) 396 kg (d) None of the above

**638.** Q x xyot; x2--!- t? (d) 20 27. (a-b)2 +1c-a)2 = 6 a2 +b2 +c2 -29 t, (a+b+C) CS -2 28. p— (b) 8.6 (c) 18 322 724B /45A 29. t? 30. 4, 6, [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q69]

(a) 1003 1005 1007 (b) 8.6

(c) 18 322 724B /45A 29. t? 30. 4, 6, (d) 20 27.

**639.** The value of (063+0-37) is 100 99 1000 (d) 999 30 In the above figure, ABCD is a parallelo- gram. p is a point on BC such that PB : PC=I : [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q72]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

**640.** The difference of maximum values of the expressions (6 + 5x — x2) and (y — 6 — y2) for any real values of x and [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q73]

(a) 16 (b) 17 (c) 18 (d) 19

**641.** A clock is started at noon. By 10 minutes past 5, through what angle, the hour hand rnoves ? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q74]

(a) 1600 (b) 1450

(c) 1500 (d) 1550 - A 32 92. minimum value of cos2x + cos2y —cos2z

**642.** The average weight of students in a class is 43 kg. Four new students are admitted to the class whose weights are 42 kg. 36•5 kg, 39 kg and 42-5 kg respectively. Now the average weight, of the students of the class is 426 kg. The number of students in the beginning was [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q75]

(a) 24 (b) —25 (c) 28 (d) —

*— SPACE FOR ROUGH WORK —*

**643.** The solution Of the inequation is ( given that x O) -I-JS -ltvs (c) [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q77]
(a) 38. Which of the points P (b) Q and R o nly (c) P and R o nly (d) P and Q o Illy

**644.** A question is given followed by two statements I and [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q79]
(a) 5 cm (b) 6 cm 59. What is PM equal to ?
(c) IO cm (d) 12 cm

**645.** XIZ is a 3-digit number, where X, Y, Z are distinct non-zero digits. The dif- ference between the two 3-digit numbers XYZ and YXZ is [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q80]
(a) 121 (b) 120 (c) 119 (d) None of the above

**646.** How many possible values exist for the sum [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q81]
(a) 121 (b) 120 (c) 119 (d) None of the above

**647.** How many numbers of the form 2n — 1 and less than 2000 are prime ? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q82]
(a) 1100 (b) 1200 (c) 1300 (d) 1400

**648.** ABCD is a square. p. Q, R, S are points on the sidesAB, BC, CD, DA respectively such that CR=DS. What is ZSPQ equal ? [CDS (II) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_ii_math_q86]
(a) 300 (b) 450 (c) 600 (d) 900

**649.** For any integer n, what is the HCF Of integers m = + I and k 9n +4 ? [CDS (I) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_i_math_q7]
(a) 4 days (b) 5 days (c) 6 days (d) ys

**650.** If k is any even positive integer, then (k2 +2k) is [CDS (I) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_i_math_q26]
(a) ivisible by (b) ivisible 8 but may not be divisible b
(c) ivisible by 4 but may not be divisible by (d) ivisible by 2 but may not be divisible by

**651.** If each time the cut was x%, then (c) x = 11 (d) x = 11.5 If 0.764 y = 1.236 x, value of [CDS (I) 2012 : Elementary Mathematics Real Test Paper Qcds_2012_i_math_q43]
(a) 0.764 (b) 0.236
(c) x = (d) x = 11.5 If 0.764 y = 1.236 x, value of

**652.** 2x3x5x7x11+1 t (c) 78, (d) 78, 2160 (ii) HCF 12 tl [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q8]
(a) 60 (b) 72 (c) 78, (d) 78, 2160

**653.** If (x+l) is the HCF of Ax2+Bx+C and Bx2 + Ax + C where A B, then the value of C is [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q18]
(a) I only (b) 2 only
(c) I and 2 only l, 2 and Y is/are 14. If (d) 2 minutes

**654.** The LCM of two numbers is 12 times their HCF. The sum Of HCF and LCM is [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q19]
(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

**655.** LCM(ab, = a LCM(b, c) [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q20]
(a) 38. Which of the points P (b) Q and R o nly (c) P and R o nly (d) P and Q o Illy

**656.** HCF(a, b) < LCM(a, b) [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q22]
(a) 38. Which of the points P (b) Q and R o nly (c) P and R o nly (d) P and Q o Illy

**657.** An employee is required to contribute of his pay to General Provident Fund. If he gets Rs. 13,500 as net pay in a month, then what is the monthly General Provident Fund contribution (assuming no other deductions) ? [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q25]
(a) Rs. 1215 (b) Rs. 1350
(c) Rs. 1500 (d) Rs. 1650 If Rs. 8400 is divided among A, B and C 28. 29. 30.

**658.** What is the remainder aner dividing the number by 9 ? S. The sum of LCM and HCF of two numbers is 536 and the difference between LCM and HCF is [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q27]
(a) 29, 17, (b) 31, 17, (c) 19, 37, (d) 23, 31,

**659.** If loglox+loglox2 = 2 log,ox+l, then what is the value ofx ? (d) 10 [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q28]
(a) only (b) k>2v5 only
(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) l—aßy

**660.** Suppose p and q are the LCM and HCF respectively of two positive numbers. If p : q = 14 I and pq 1134, then what is the difference between the two numbers? [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q29]
(a) 27 (b) 35
(c) 45 (d) nnot be determined due insufficient data

**661.** What is the HCF of 236-1 and 245-1 ? [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q30]
(a) 1023 (b) 512 (c) 511 (d) 255

**662.** If 10500 and y what is logy equal to ? [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q46]
(a) 2500 (b) 5625 (c) 5625 (d) 2500

**663.** If 5-t¯l then what is the value of x ? (b) logr02 (c) log105 (d) 210g105 [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q47]
(a) 30 (b) 45 (c) 60 (d) 75

*— SPACE FOR ROUGH WORK —*

**664.** If the LCMisx (a 2-digit number), then how many values of x are possible ? [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q48]

(a) Only one (b) Only two (c) Only three (d) More than three

**665.** p q (b) 200 (c) 2 05 (d) 210 2 3 2. 21. n Mt 22.3, 9, 27, 81, 243, 729, [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q67]

(a) 95 (b) 200

(c) 2 (d) 210 2 3 2. 21. n Mt 22.3, 9, 27, 81, 243, 729,

**666.** What •is the HCF 4x3 + 3x2y—9xy2 uy3 [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q69]

(a) x—2y of and third side in cm? (b) g, 2v'fi)

(c) (442 (d) (44ä, 21,110)

**667.** How many digits are there in (54)10 ? (Given that log102 — 0-301 and log,03 0477) [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q77]

(a) 16 (b) 18 (c) 19 (d) 27 3 (b) (c)

**668.** qR p q (p (HCF) G, t, 1. (HCF) 11. p, (p-q) (HCF) (b) 11 (c) 1 +11 ä (d) 11 23. p q (HCF) (LCM) 44 4620 I p [CDS (II) 2011 : Elementary Mathematics Real Test Paper Qcds_2011_ii_math_q97]

(a) 770 (b) 11 (c) 1 +11 ä (d) 11 23. p q

**669.** Question : Is p2 + q2 + q Odd, where p , q are positive integers ? [CDS (II) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_ii_math_q5]

(a) 24 (b) 36 (c) 48 (d) 60

**670.** What is the value of p when x —1, y = 1, [CDS (II) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_ii_math_q7]

(a) 100 (b) 125 (c) 135 (d) 140

**671.** What is the remainder when 3255 is divided by 28 ? (b) 11 (c) 24 (d) [CDS (II) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_ii_math_q8]

(a) nM — k (b) n (c) 81 (d) 105

**672.** What is the difference between their present ages? [CDS (II) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_ii_math_q9]

(a) 20 years (b) 22 years

(c) 24 years (d) 25 years A mixture contains milk and water in the ratio 5 :

**673.** The number 199 can be written as m2 — n2, where m, n are natural numbers (m > n). What is the value of mn ? [CDS (II) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_ii_math_q11]

(a) 9900 (b) 9800

(c) 9701 (d) nnot be uniquely determined

**674.** Let S — 7b + IIC +13d, where a, b, c and d are natural numbers. What is the number of distinct remainders of S when it is divided by 10 ? (d) More than [CDS (II) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_ii_math_q12]

(a) 1100 (b) 1200 (c) 1300 (d) 1400

**675.** ZTftp+q+r=OtÄ (d) z3 (k2 — 5k + 4)x2 + (k 2 — 3k — + b2+C2 = c2+a2 ¯ a2+b [CDS (II) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_ii_math_q36]

(a) 2b2 + b2c2 + c2a2 (b) 2 (c) 3 (d) z3

**676.** What is the remainder when x5 — 5x2 + 125 is divided by x +5? (b) 125 (c) -3125 (d) 3125 IA 16 •and ab(x2 — —b2)? (a2x2 bx)2 (a2x2 of [CDS (II) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_ii_math_q53]

(a) 125 (b) -3125

(c) 3125 IA 46. (d) e 160 only e 240 only e 160 or 240 Neither e 160 nor 240

**677.** Ihe number of values of x satisfying 100 x + — > 50 , where x is a natural number less than or equal to 100 is [CDS (II) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_ii_math_q54]

(a) 51 (b) 53 (c) 55 (d) 57

**678.** The seven digit .number 876p37q is divisible by [CDS (II) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_ii_math_q57]

(a) I only (b) 2 only

(c) I and 2 only I, 2 and Y is/are 14. If (d) 2 minutes

**679.** Consider thc following in respect of natural numbers a, b and c : [CDS (II) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_ii_math_q60]

(a) 38. Which of the points P (b) Q and R o nly (c) P and R o nly (d) P and Q o Illy

**680.** What can be said about the expansion of 212n—64n, where n is a positive integer ? [CDS (I) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_i_math_q1]

(a) Last digit is (b) Last digit is (c) Last digit is (d) Last two digits are zero

**681.** x, I X, 252 t, y, 308 z, 198 [CDS (I) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_i_math_q6]

(a) 26 fiaz (b) 42 Raz

(c) 45 (d) 46 fraz 1 $24 18. — , — 3 6 9'

**682.** If the highest common factor of two positive integers is 24 then their least common multiple cannot be [CDS (I) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_i_math_q16]

(a) 72 (b) 216 (c) 372 (d) 600

**683.** If the sum of the digits of a number 10" — 1, where n is a natural number, is equal to 3798, then what is the value of n? [CDS (I) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_i_math_q23]

(a) 421 (b) 422 (c) 423 (d) 424

**684.** What is the value of x 8) if (10097 + 10054 + x + 1) leaves a remainder O when divided by 9 ? A - [CDS (I) 2010 : Elementary Mathematics Real Test Paper Qcds_2010_i_math_q39]

(a) nM — k (b) n (c) 81 (d) 105

*— SPACE FOR ROUGH WORK —*

**685.** Consider those numbers between 300 and 400 such that when each number is divi&d by 6, 9 and 12. it leaves 4 as remainder in each case. What is the sum of the numbers ? [CDS (I) 2009 : Elementary Mathematics Real Test Paper Qcds_2009_i_math_q2]

(a) 692 (b) 764 (c) 1080 (d) 1092 - A

**686.** If3X+27(3 ) -x = 12, then what is the value of x? [CDS (I) 2009 : Elementary Mathematics Real Test Paper Qcds_2009_i_math_q6]

(a) 1 only (b) 2 only (c) 1 or (d) O or I

**687.** What is 0f50% of 500 ? [CDS (I) 2009 : Elementary Mathematics Real Test Paper Qcds_2009_i_math_q80]

(a) 12.5 (b) 25 (c) 1.25 (d) 6.25

## SECTION : GEOMETRY: CIRCLES, CHORDS & TANGENTS (9 QUESTIONS)

**688.** Smaller lead shots are to be prepared by using the material of a spherical lead shot of radius I cm. Some possibilities are listed in the statements given below: . I. The material is just sufficient to prepare 8 shots each of radius cm. [CDS (II) 2016 Q99]

(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ? (b) 9th

(c) 15m (d) I •5m

**689.** The opposite angles of a cyclic quadrilateral are supplementary. Angle subtended by an arc at the centre is double the angle subtended by it at any point On the remaining part Of the circle. Which one of the following is correct in respect of the above statements ? [CDS (II) 2023 Q85]

(a) Statement- I statement-2 (b) Statement-2 statement-I

(c) Statement-I statement-2 (d) Neither statement-I statement-2 nor statement-2 statement-I (a) (c) (d)

**690.** A piu of wim of 33 is into an rc of a ofradius 14 is the angle by am at ca&e of circle ? (c) 1350 (d) 1500 C - SDHY T-MTÄ [CDS (II) 2023 Q90]

(a) 32% (b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?

(c) cm2 (d) 50

**691.** A shot of radius 0-75 cm and a second shot of radius 0-8 cm can be prepared from the available materiaL Which of the above statements is/ are correct ? [CDS (II) 2016 Q100]

(a) 1 Only (b) 2 Only (c) Both I and 2 (d) Neither nor 2

**692.** If two chords AB and AC Of a circle are equal, then the centre Of the circle lies on the angle bisector Of angle CAB. If two concentric circles are intersected by a line at A, B, C and D respectively, then AC BD. Which Of the above statements is/are correct? [CDS (II) 2017 Q31]

(a) I Only (b) 2 only (c) Both I and 2 (d) Neither I nor 2

**693.** The average of the ages of 15 students in a class is 19 years. When 5 new students are admitted to the class, the average age of the class becomes 18-5 years. What is the average age of the 5 newly admitted students ? [CDS Math PYQ Chapter ch_10]

(a) 17 years (b) 17*5 years (c) 18 years (d) 18-5 years

**694.** A water tank, open at the top, is hemi- spherical at the bottom and cylindrical above it. Ihe radius is 12m and the capacity is 3312rt m3. The ratio of the surface areas of the spherical and cylindrical portions is - A [CDS Math PYQ Chapter ch_10]

(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

**695.** The cost of painting a spherical vessel of diameter 14 cm is Rs. 8008. What is the cost Of painting per square centimetre ? [CDS Math PYQ Chapter ch_10]

(a) Rs. 8 (b) Rs. 9 (c) Rs. 13 (d) Rs. 14

**696.** Let O be the centre of a circle. Let A, B and C lie on the circle such that LABC= 1530. What is the value of LAOC ? 21 [CDS Math PYQ Chapter ch_10]

(a) 540 (b) 520 (c) 390 (d) 270

## SECTION : MENSURATION 3D: SURFACE AREAS & VOLUMES (3 QUESTIONS)

**697.** A cylinder having base of circumference 60 cm is rolling without sliding at a rate of 5 rounds per second. How much dis- tance will the cylinder roll in 5 seconds ? [CDS (II) 2017 Q53]

(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ? (b) 9th

(c) 15m (d) I •5m

**698.** What is volume of the frustum of a cone with height h and radii r, , ? [CDS (II) 2025 Q46]

(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ? (b) 9th

(c) 15m (d) I •5m

**699.** 20 What is the difference in radii of the ends of the frustum ? (b) (d) None of the above What is the volume ofthe frustum ? [CDS (II) 2025 Q47]

(a) 12 (b) 13 If 196x4 = x6, then x3 is equal to which one of the following ?

(c) x'/ 14 (d) 14xa

## SECTION : STATISTICS & DATA INTERPRETATION (4 QUESTIONS)

**700.** What is the weighted mean of first IO natural numbers whose weights are equal to the goffesponding numlYr ? (d) 4-5 [CDS (I) 2026 Q53]

(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ? (b) 9th

(c) 15m (d) I •5m

**701.** Marks of students who appeared for a test of 100 marks. Collar sizes of200 shirts sold in a week. Monthly incomes of250 employees Of a factory. For which of the above data, mode is a suitable measure of central tendency ? [CDS (I) 2026 Q65]

(a) I and 2 only (b) 2 Only (c) I and 3 only (d) 1 , 2 and 3

*— SPACE FOR ROUGH WORK —*

**702.** The age of a woman is a two-digit integer. On reversing this integer, the new integer is the age of her husband who is elder to her. The difference between their ages is one-eleventh of their sum. What is the difference between their ages ? [CDS Math PYQ Chapter ch_13]

(a) 8 years (b) 9 years (c) 10 years (d) 11 years

**703.** If the mean age of combined group Of boys and girls is 18 years and-the mean Of age of boys is 20 and that of girls is 16; then what is the percentage of boys in the group ? [CDS Math PYQ Chapter ch_13]

(a) 60 (b) 50 (c) 45 (d) 40 - A 12

## SECTION : MENSURATION 2D: AREAS & PERIMETERS (4 QUESTIONS)

**704.** The perimeter of the triangle is Select the answer using the code given below : [CDS (II) 2017 Q22]

(a) I only (b) 11 only (c) Both I and 11 (d) Neither I nor 11 23

**705.** The perimeter of a triangle is greater than the sum of the lengths of its three medians. Which of the statemepts given above is/are correct ? [CDS (I) 2022 Q70]

(a) only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2 - A 16

**706.** The area of a trapezium is 336cm2. If its parallel sides are in the ratio 5 : 7 and the perpendicular distance between them is 14 cm, then the smaller of the parallel sides is [CDS Math PYQ Chapter ch_11]

(a) 20 cm (b) 22 cm (c) 24 cm (d) 26 cm

**707.** A rhombus is formed by joining midpoints of thc sides of a rectangle in the suitable order. If the area of the rhombus is 2 square units, then the area Of the rectangle is [CDS Math PYQ Chapter ch_11]

(a) 242 square units (b) 4 square units (c) 40 square units (d) 8 square units

*— SPACE FOR ROUGH WORK —*